

Machine Tools

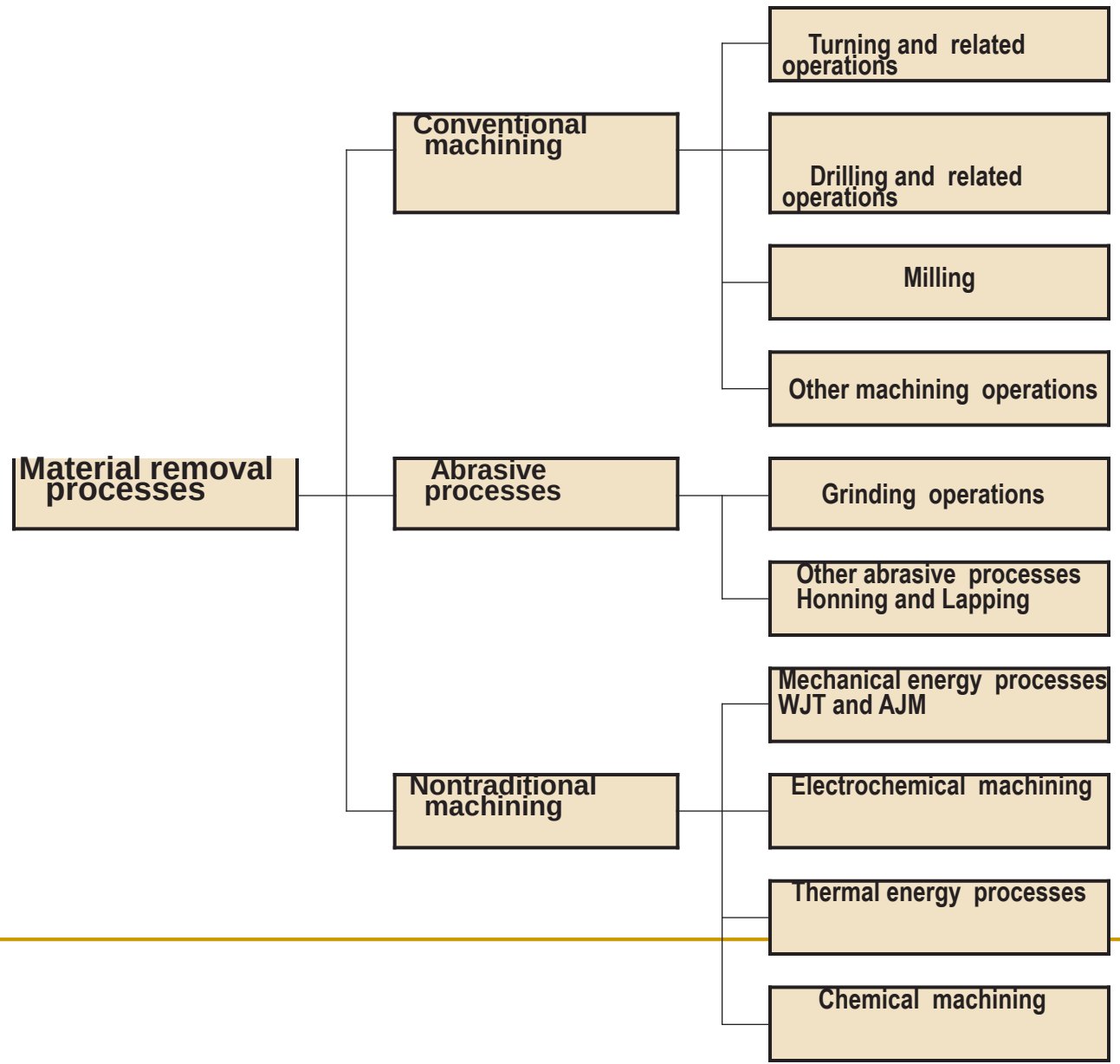


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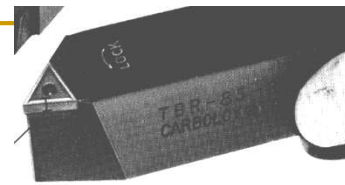
Categories of Manufacturing Processes

- **Casting & Molding**: Liquid material is poured into a mold, such as sand casting, die casting, or plastic injection molding.
- **Forming & Metalworking**: Raw materials are shaped using mechanical force, including forging, rolling, and extrusion.
- **Machining (Material Removal)**: Unwanted material is removed to achieve precise dimensions via drilling, milling, and turning.
- **Joining & Assembly**: Components are permanently or temporarily connected, such as welding, brazing, soldering, and fastening.
- **Additive Manufacturing (3D Printing)**: Building objects layer by layer, suitable for complex, customized parts.
- **Surface Finishing**: Processes like coating, painting, polishing, and heat treating to improve surface properties.

Material Removal Processes (Machining)



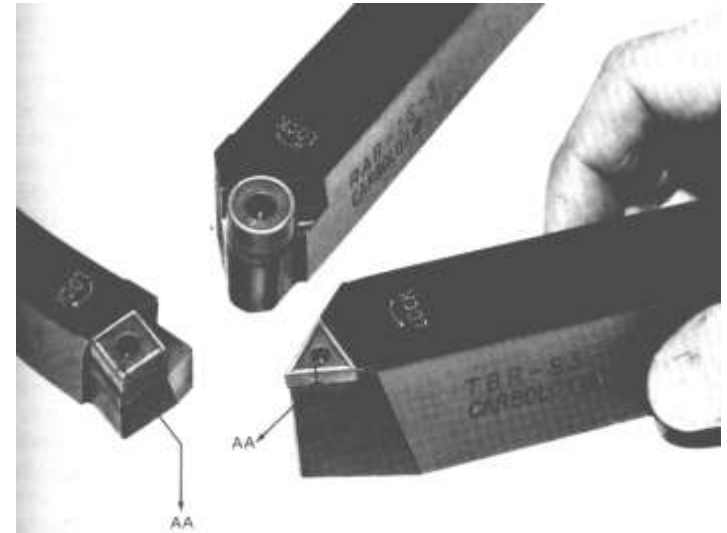
Cutting Tool



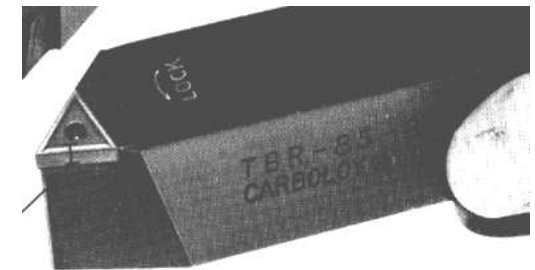
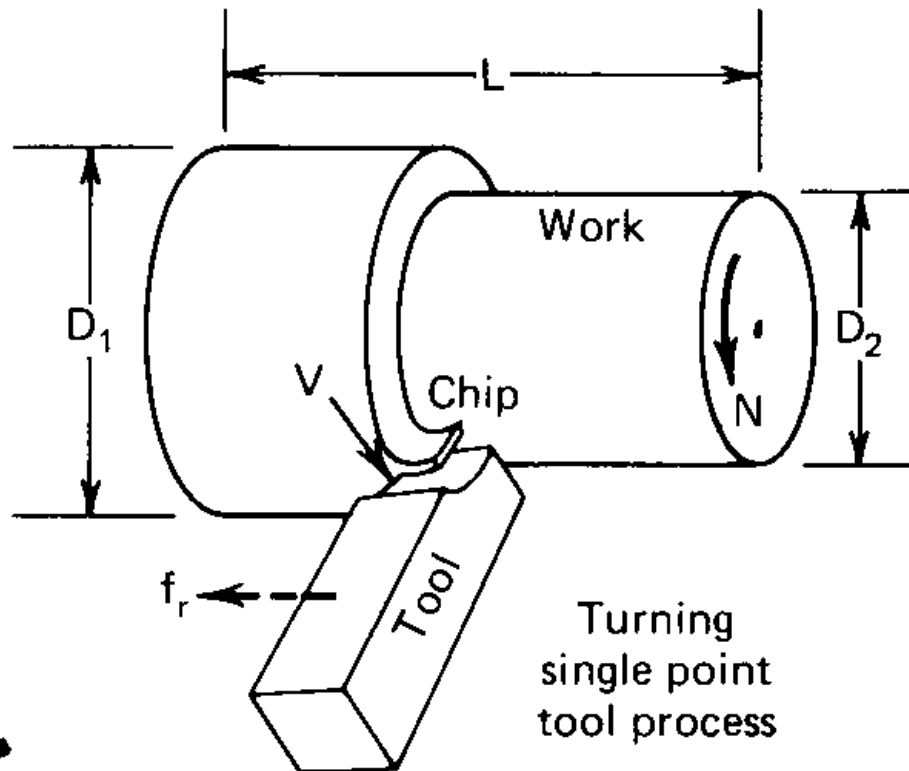
- In the context of metalworking, a **cutting tool**, is any tool that is used to remove metal from the work piece in form of chips.
- It frequently refers to a tool bit.
- Tool material should be harder than the material which is to be cut
- They must be able to withstand the heat generated in the metal cutting process.
- They also must have a specific geometry, designed so that the cutting edge can contact the work piece without the rest of the tool dragging on its surface.

Classification of Cutting tools

- single point cutting tool
- multiple point cutting tool.

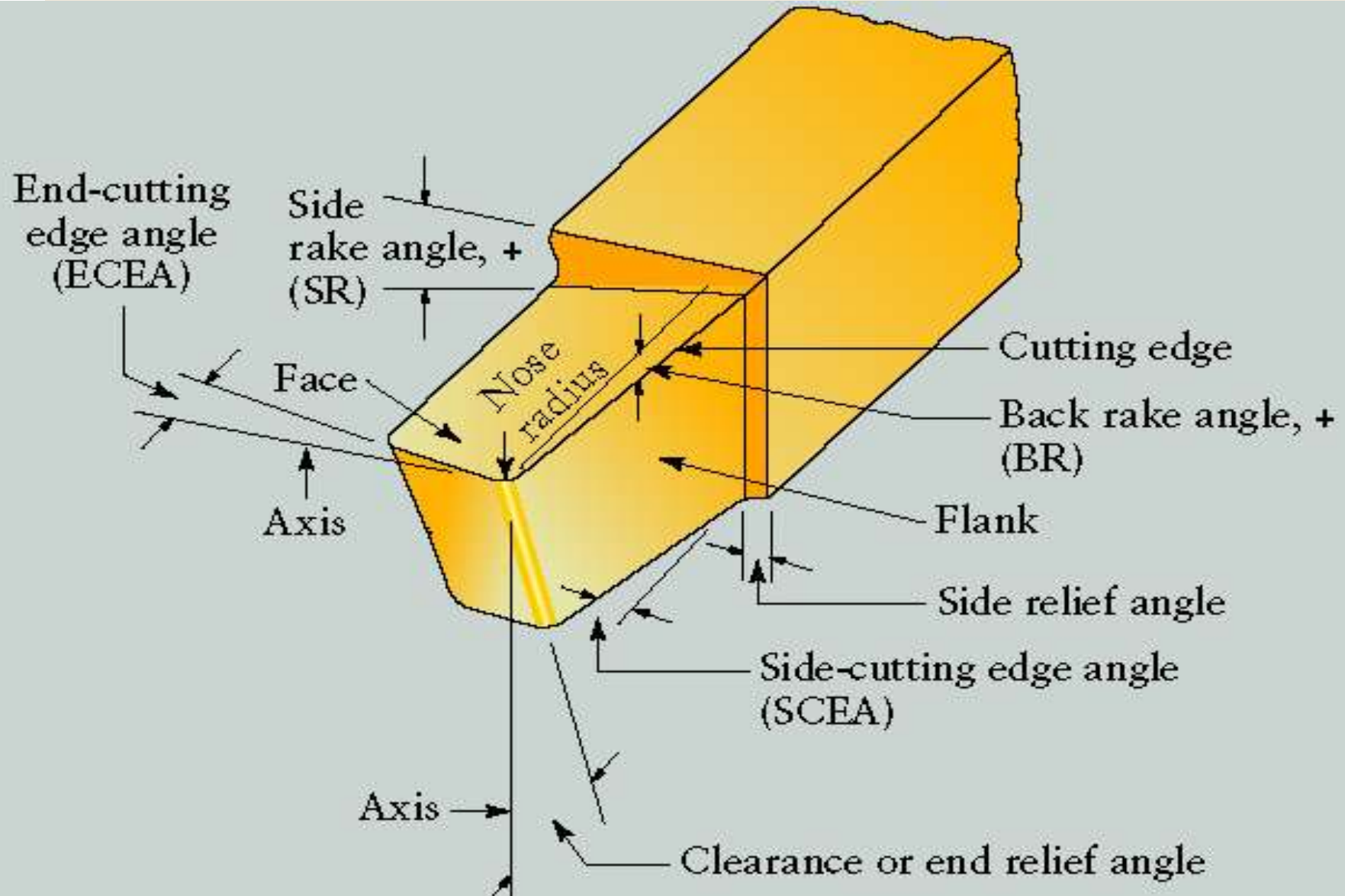


A single-point cutting tool

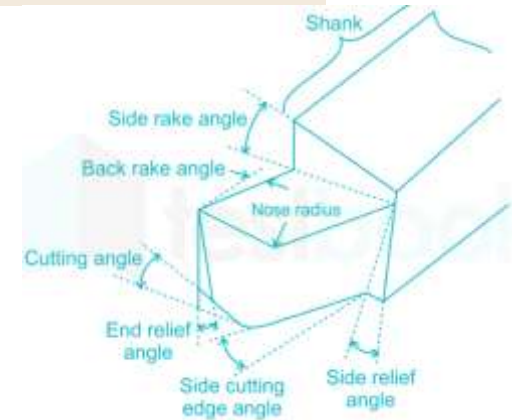


Cutting speed = $\pi DN/60$ mm per second

Single Point Cutting Tool geometry

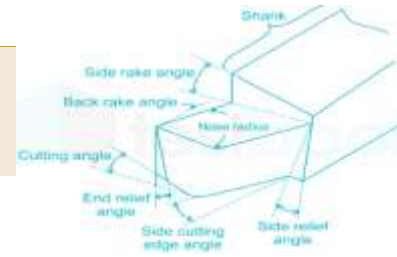


Know the Single Point Cutting Tool



- **Shank:** Main body of tool, it is part of tool which is gripped in tool holder
- **Face:** Top surface of tool b/w shank and point of tool. Chips flow along this surface
- **Flank:** Portion of tool which faces the work. It is surface adjacent to & below the cutting edge when tool lies in a horizontal position.
- **Base:** Bearing surface of tool on which it is held in a tool holder.
- **Nose radius:** Cutting tip, which carries a sharp cutting point. Nose provided with radius to enable greater strength, increase tool life & surface life.
Typical Value : 0.4 mm – 1.6 mm

Single point Cutting-Tool Terms

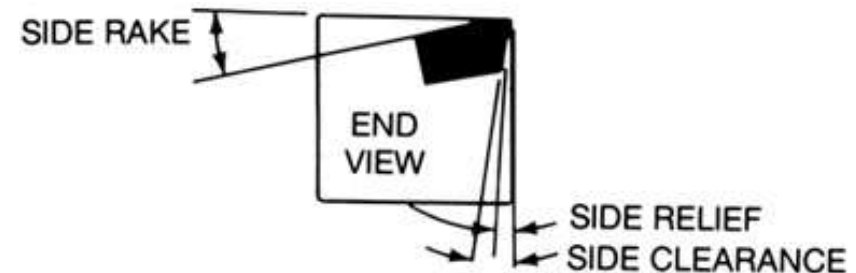


Relief or Clearance angle:

- ❑ Ground on the end and side faces of a tool to prevent it from rubbing on the work piece.
- ❑ To enable only the cutting edge to touch the work piece.

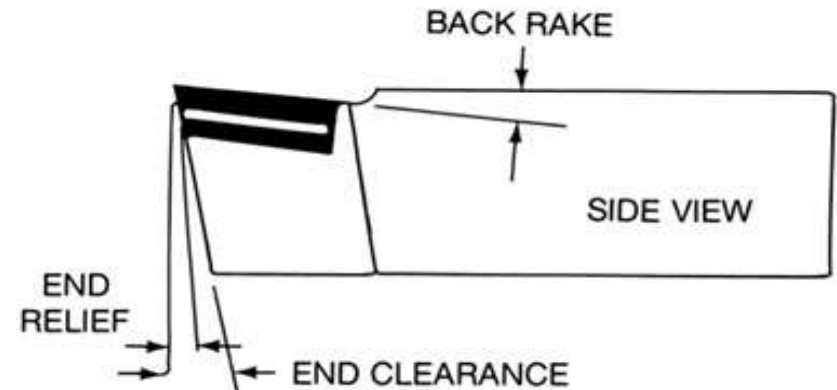
Side Relief angle:

- Angle ground directly below the cutting edge on the flank of the tool

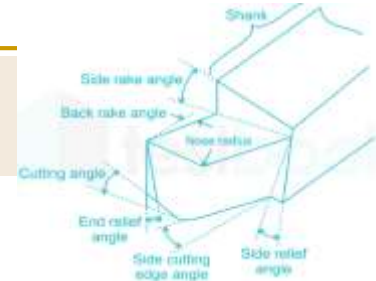


End Relief angle:

- Angle ground from the nose of the tool



Cutting-Tool Terms

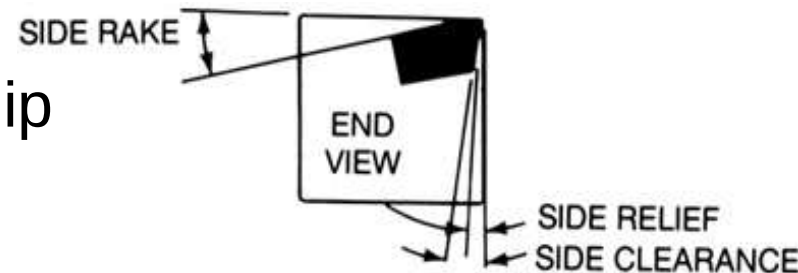


Rake angle:

- Ground on a tool to provide a smooth flow of the chip over the tool so as to move it away from the work piece

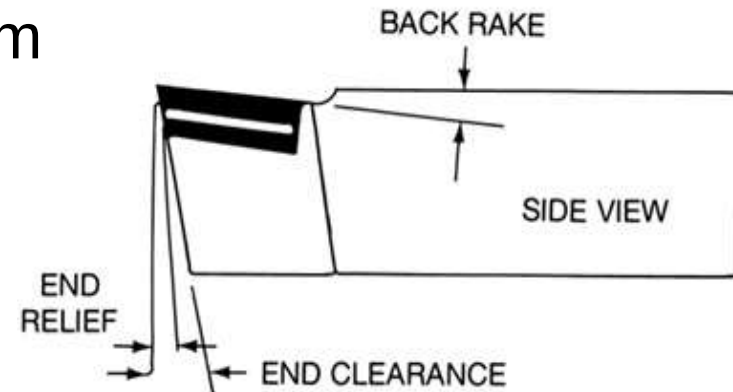
Back Rake angle

- Ground on the face of the tool
- Influences the angle at which chip leaves the nose of the tool
- Generally $8 - 10^\circ$

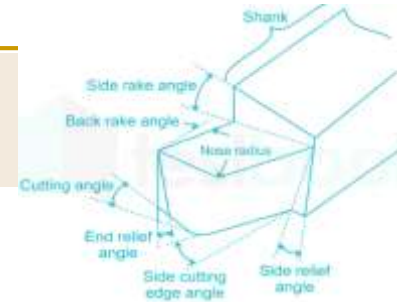


Side Rake angle

- Ground on the tool face away from the cutting edge
- Influences the angle at which the chip leaves the work piece
- A lathe tool has 14° side rake.



Back Rake Angle



- Angle formed between top face of tool and top of tool shank

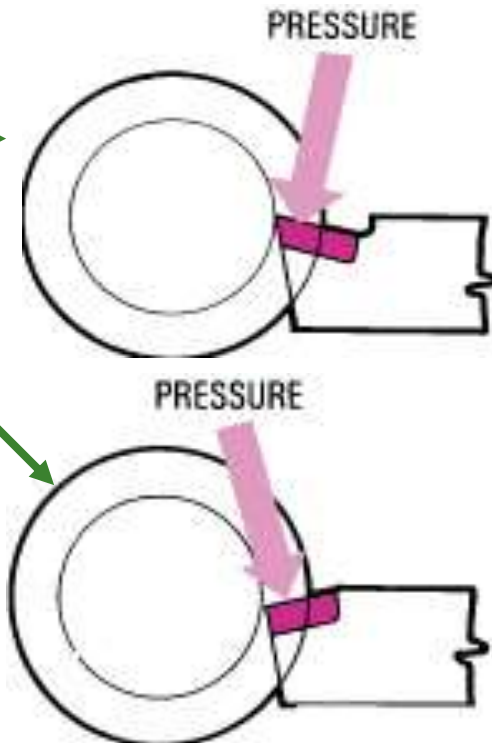
- Positive

- Top face slopes downward away from point

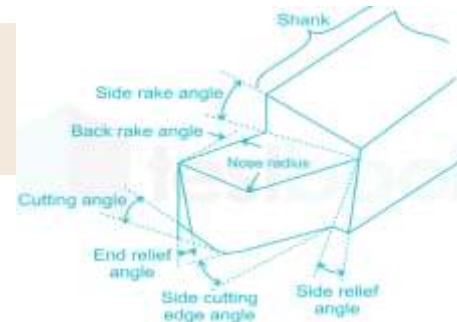
- Negative

- Top face slopes upward away from point

- Neutral



Cutting-Tool Terms

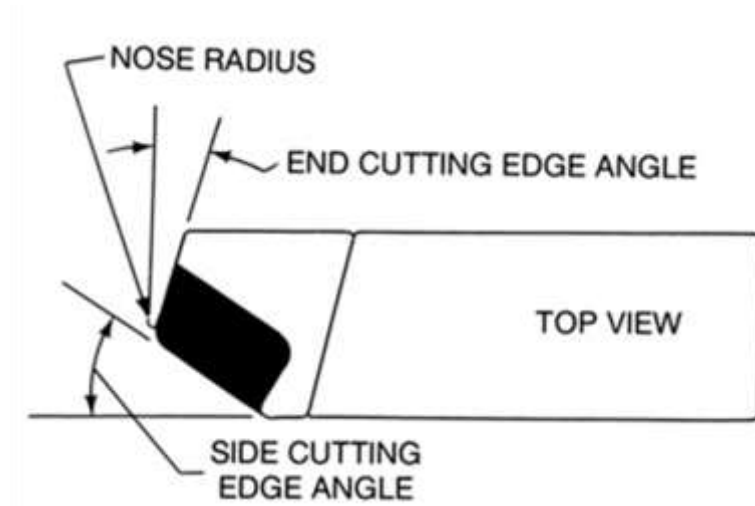


Nose Radius:

- **Rounded tip on the point of the tool**

Functions:

- Strengthens finishing point of tool
- Improves surface finish on work
- Should be twice amount of feed per revolution
 - Too large – chatter; too small – weakens point

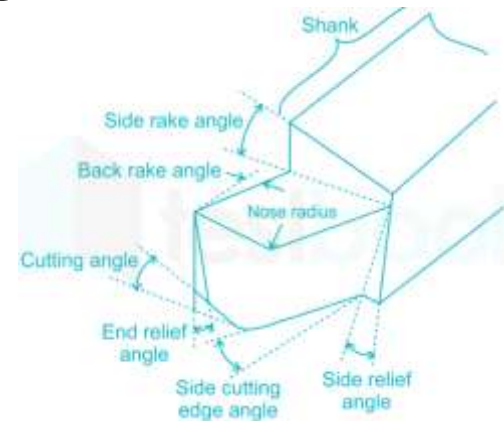


Tool Signature

- Convenient way to specify tool angles by use of a standardized abbreviated system is known as tool signature or tool nomenclature.
- The seven elements that comprise the signature of a single point cutting tool can be stated in the following order:

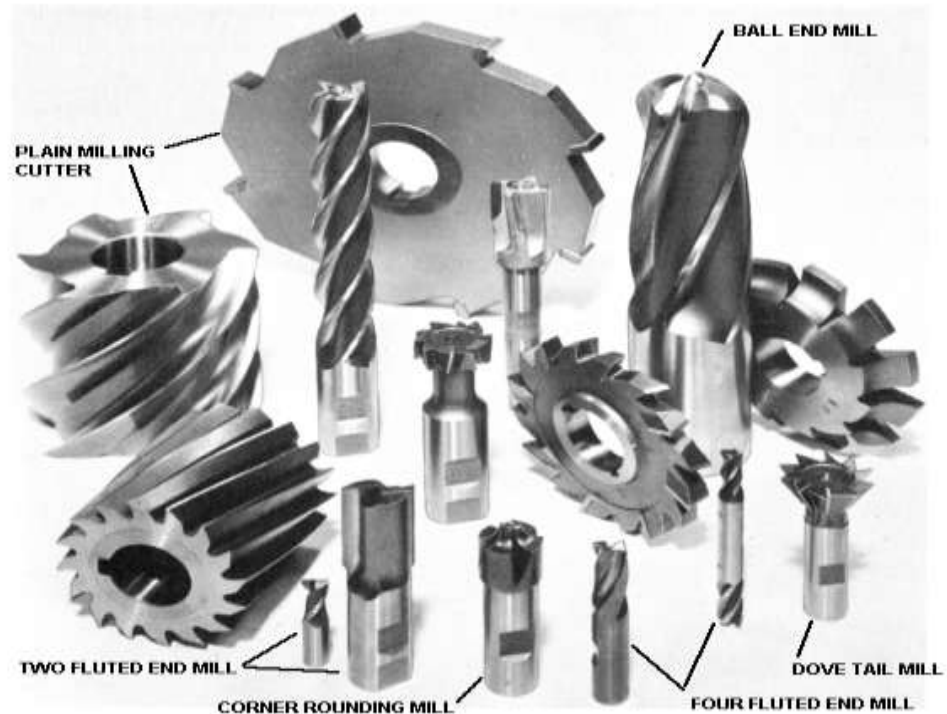
Tool signature 0-7-6-8-15-16-0.8

1. Back rake angle (0°)
2. Side rake angle (7°)
3. End relief angle (6°)
4. Side relief angle (8°)
5. End cutting edge angle (15°)
6. Side cutting edge angle (16°)
7. Nose radius (0.8 mm)



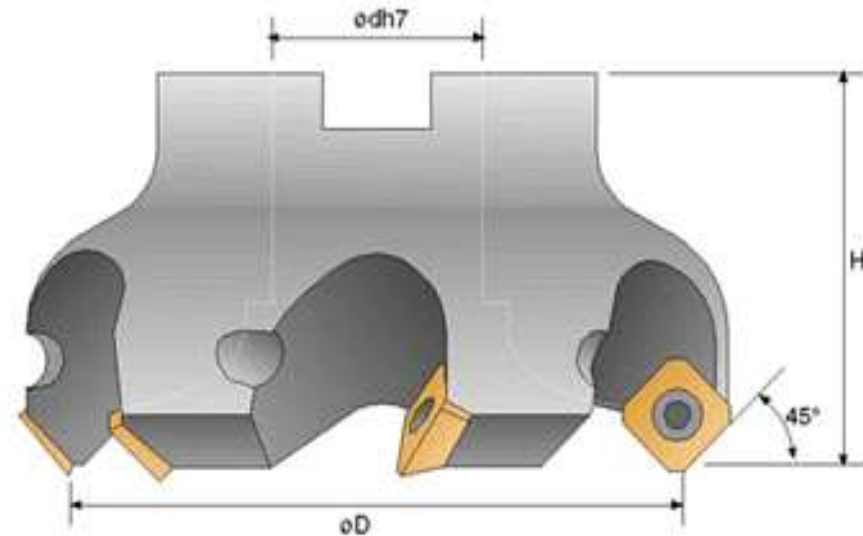
Multiple-point cutting tools

- have two or more cutting edges.
- Example:- milling cutters, drills, and broaches.

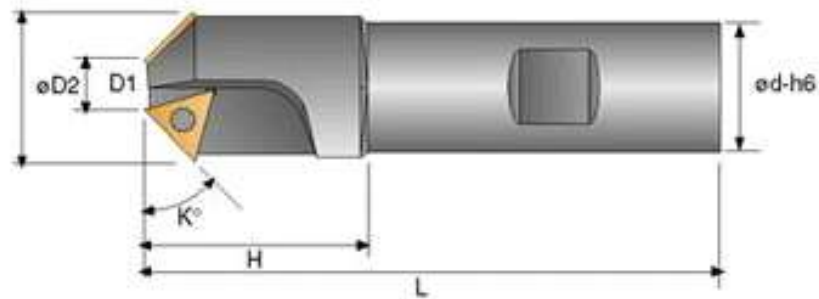


Milling Tool Geometry

Face cutter



Chamfering cutter



Cutting Tool Materials

- Tool Steels
- High Speed Steel (HSS)
- Carbides
- Coated Carbide
- Ceramics and cermet
- TiN Coated High-Speed Steel
- CBN- Carbon Boron Nitride
- Diamond tool

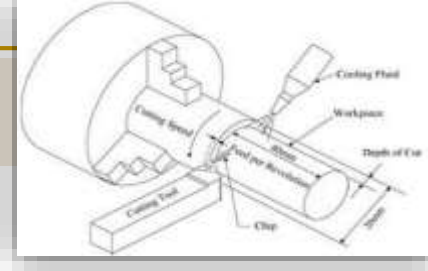
Machine tools

- ❑ Lathe turning
- ❑ Drilling
- ❑ Milling
- ❑ Shaper
- ❑ Planer
- ❑ Grinding
- ❑ Broaching, sawing, filing
- ❑ Nontraditional machining process (ex EDM, CNC)

Basic functional principles of machine tool operations:

- Machine Tools produce desired geometrical surfaces on solid bodies (preformed blanks) and for that they are basically comprised of;
 - ⑩ Devices for firmly holding the tool and work
 - ⑩ Drives for providing power and motions to the tool and work
 - ⑩ Kinematic system to transmit motion and power from the sources to the tool-work
 - ⑩ Automation and control systems
 - ⑩ Structural body to support and accommodate those systems with sufficient strength and rigidity.

Definition of Machining



- Machining is an essential process of finishing by which jobs are produced to the desired dimensions and surface finish by gradually removing the excess material from the preformed blank .
- To perform the operation, **relative motion is required between the tool and the work.**
- For material removal by machining, the work and the tool need relative movements and those motions and required power are derived from the power source(s)
- In the turning process, the speed of chuck/spindle is set at certain **RPM (Rotations per minute)** to cut /machine surfaces of workpiece or blank.
- The cutting tool is set at a certain **depth of cut (mm)** and travels to the left with a certain velocity as the workpiece rotates.
- The **feed, or feed rate**, is the distance the tool travels horizontally per unit revolution of the workpiece (mm/rev).
- This movement of the tool produces a **chip**, which moves up the face of the tool.

Purpose of Machining:

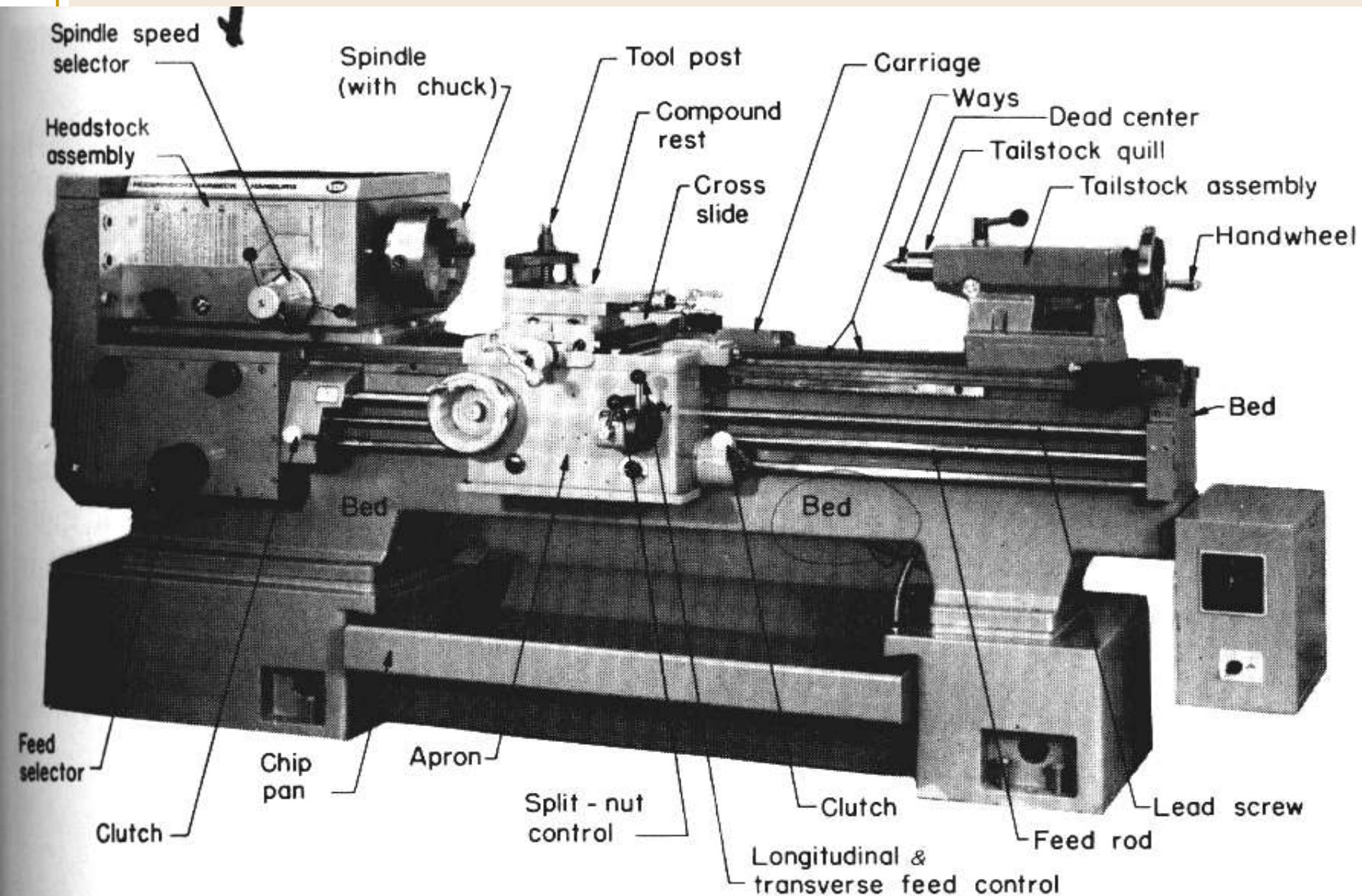
- Most of the engineering components such as **gears, bearings, clutches, tools, screws and nuts** etc. need dimensional and form accuracy and good surface finish for serving their purposes.
- Preforming like **casting, forging etc. generally cannot provide the desired accuracy and finish.**
- For that such preformed parts, called blanks, need semi-finishing and finishing and it is done by machining and grinding.
- Grinding is also basically a machining process.
- Machining to high accuracy and finish essentially enables a product
 - ⑩ Fulfill its functional requirements
 - ⑩ Improve its performance
 - ⑩ prolong its service.

Basic functions of Machine Tools

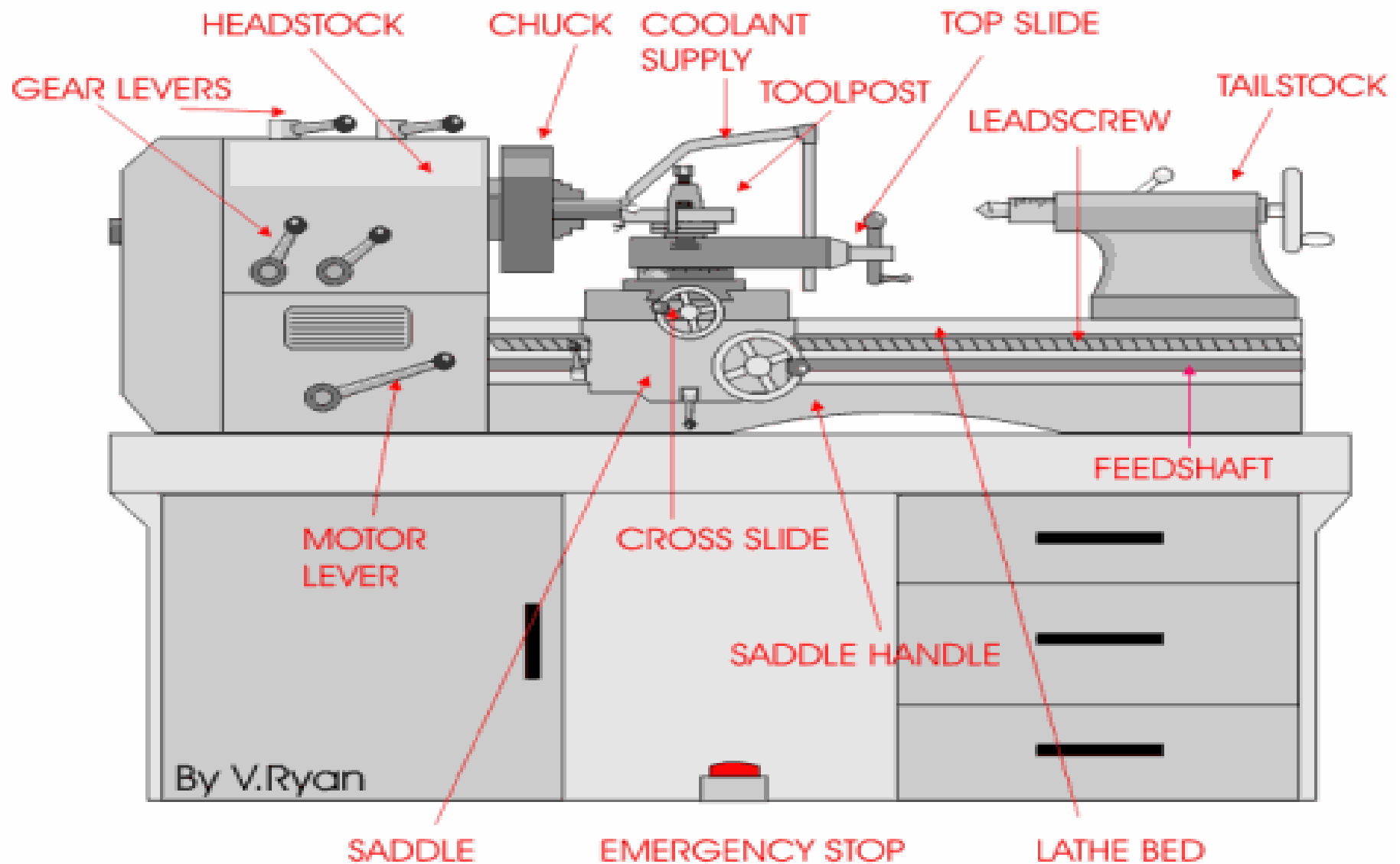
- Machine Tools basically produce geometrical surfaces like flat, cylindrical or any contour on the preformed blanks by machining work with the help of cutting tools.
- The physical functions of a Machine Tool in machining are:
 - ⑩ Firmly holding the blank and the tool
 - ⑩ Transmit motions to the tool and the blank
 - ⑩ provide power to the tool-work pair for the machining action.
 - ⑩ **Control of the machining parameters**, i.e., speed, feed and depth of cut.



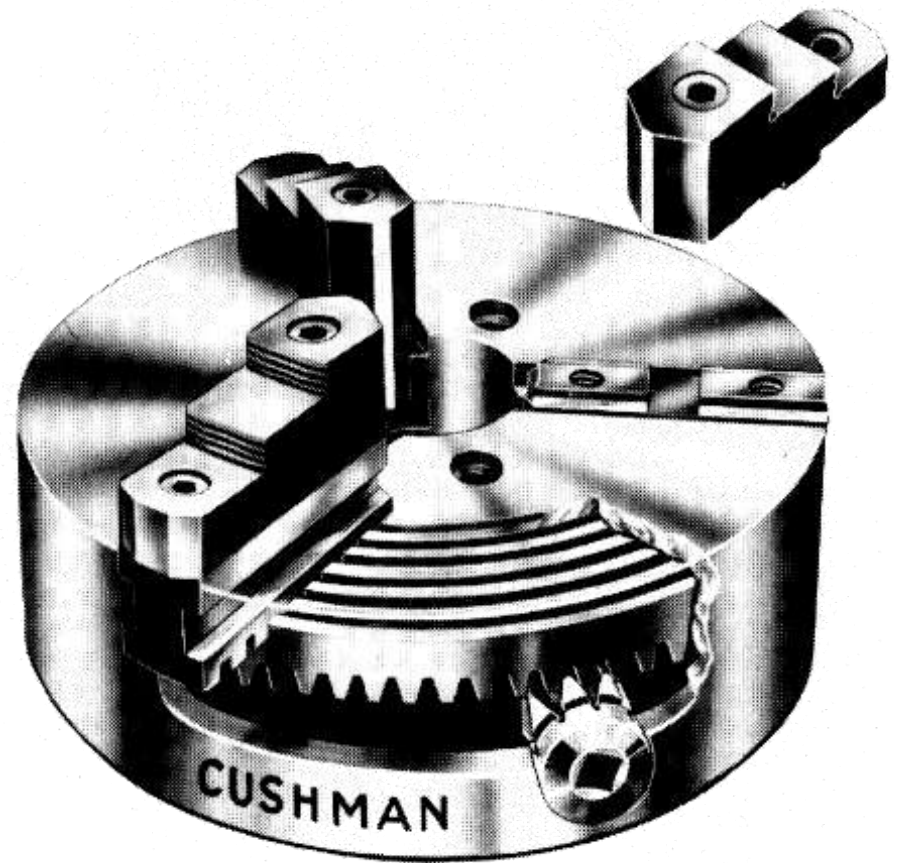
Lathe machine...



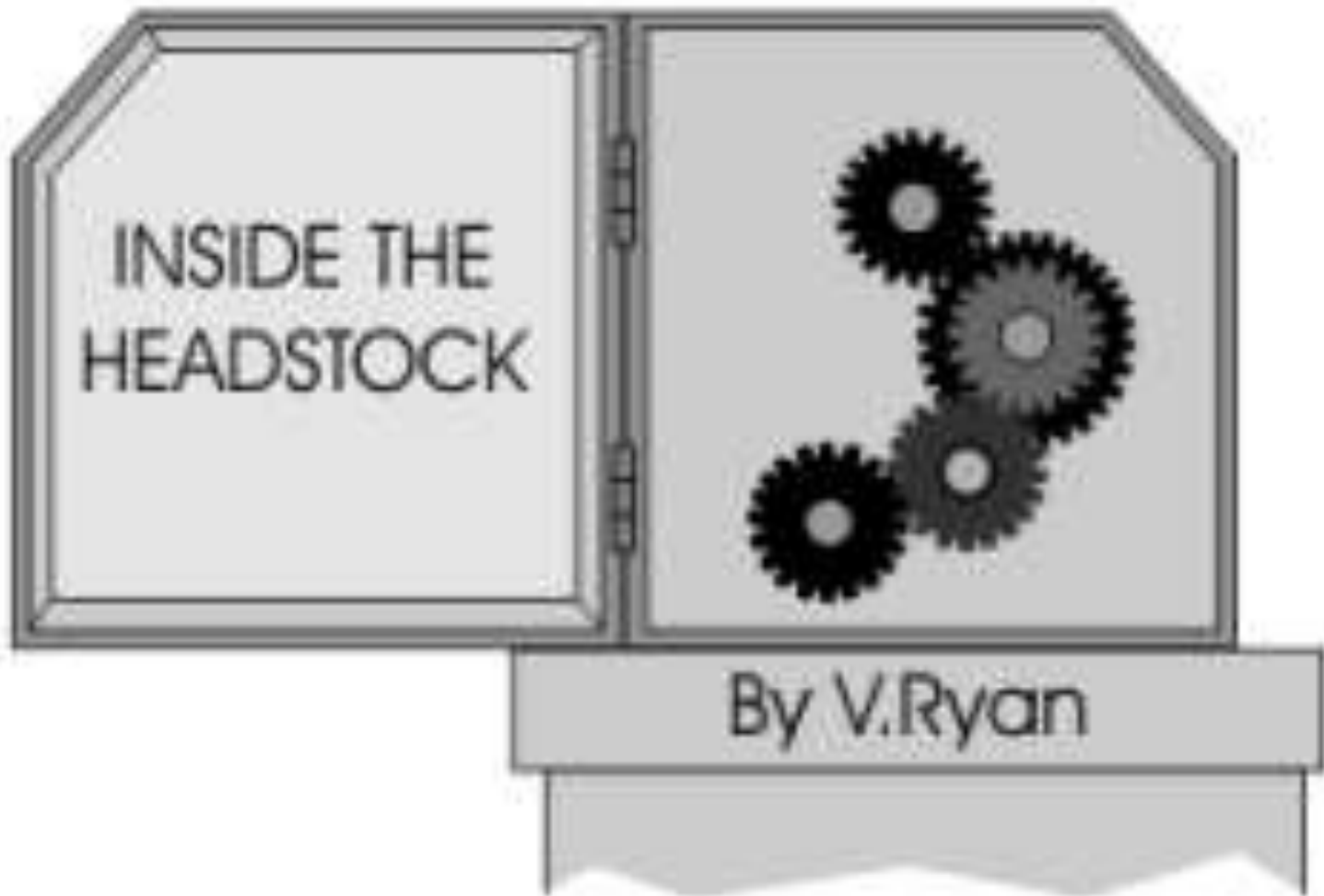
Lathe



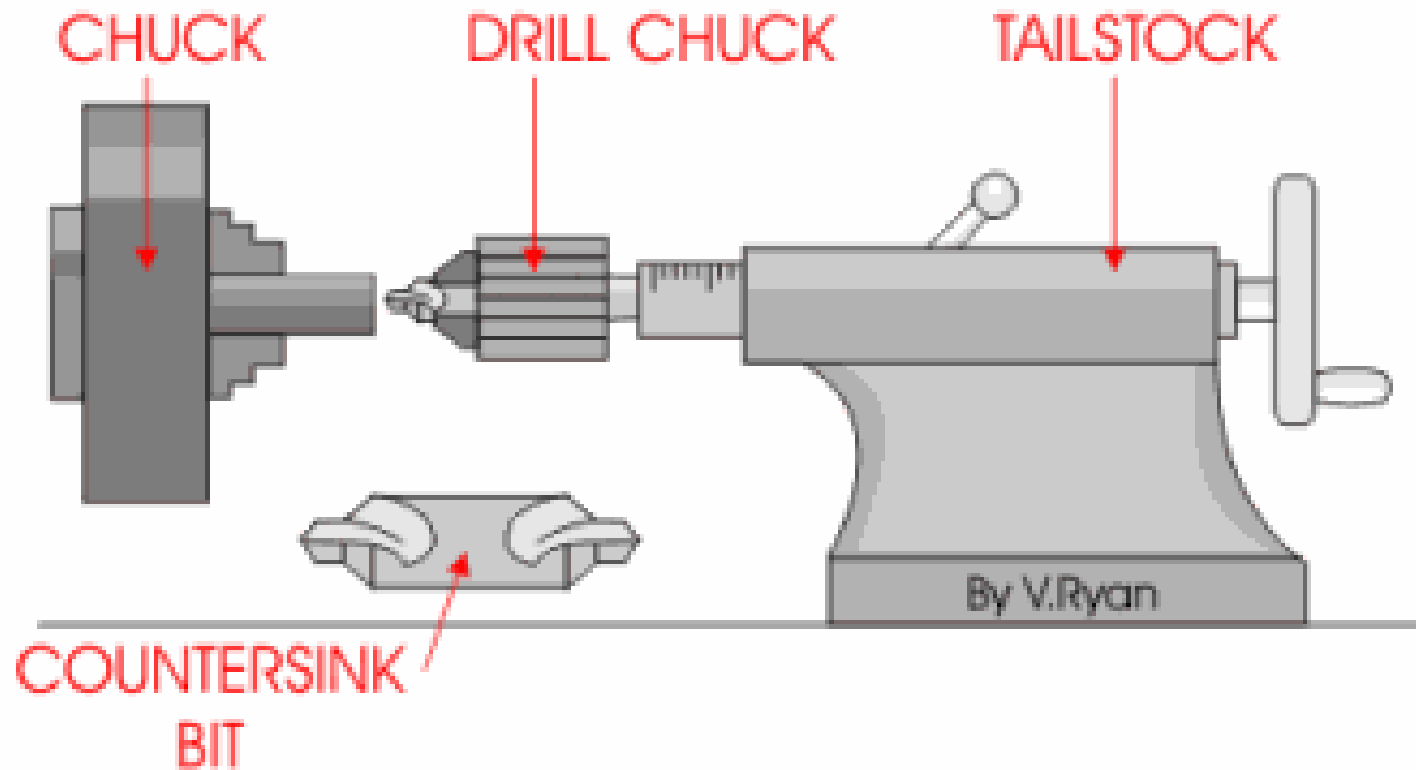
Chuck



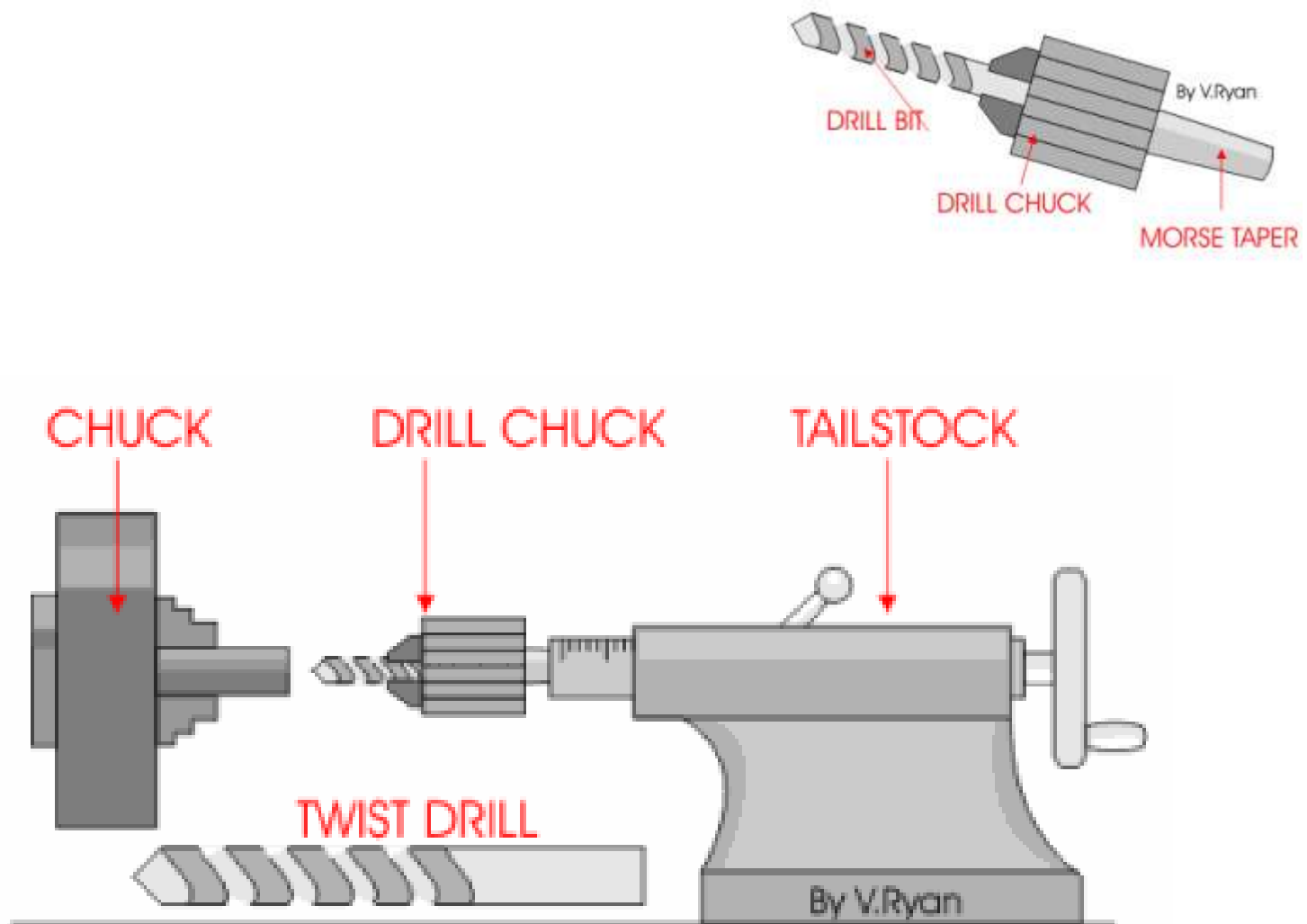
Gear System



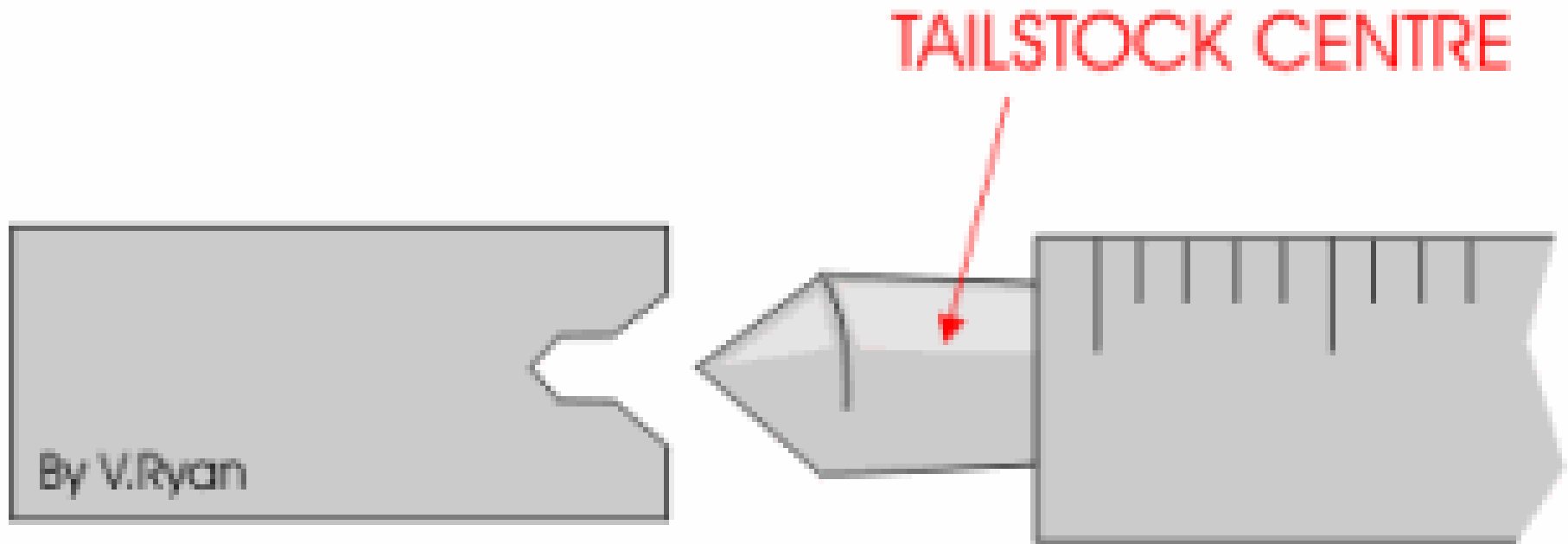
Tailstock



Drilling in lathe machine



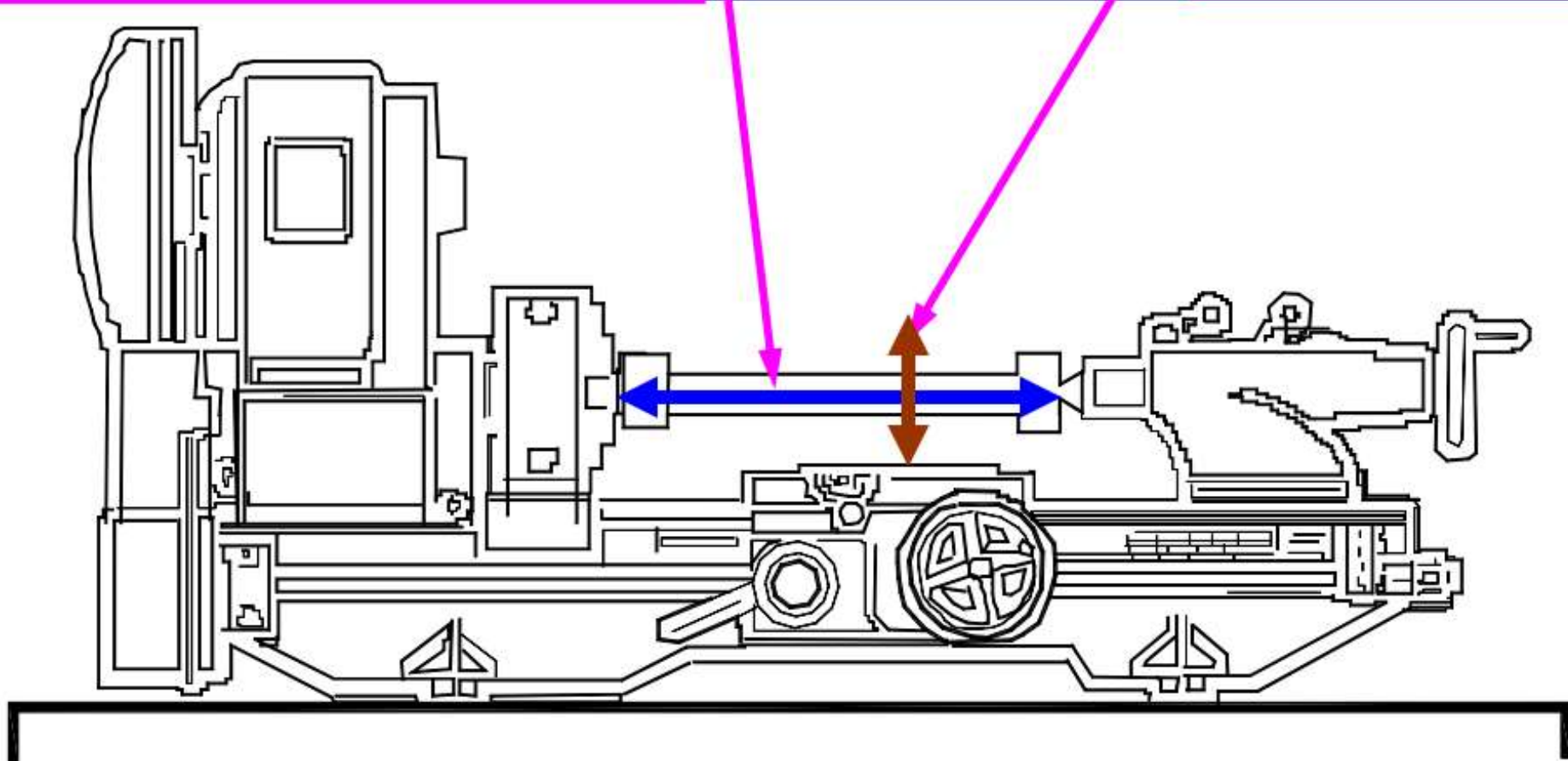
Long Job is supported by tailstock



Size of Lathe machine

Workpiece Length

Swing

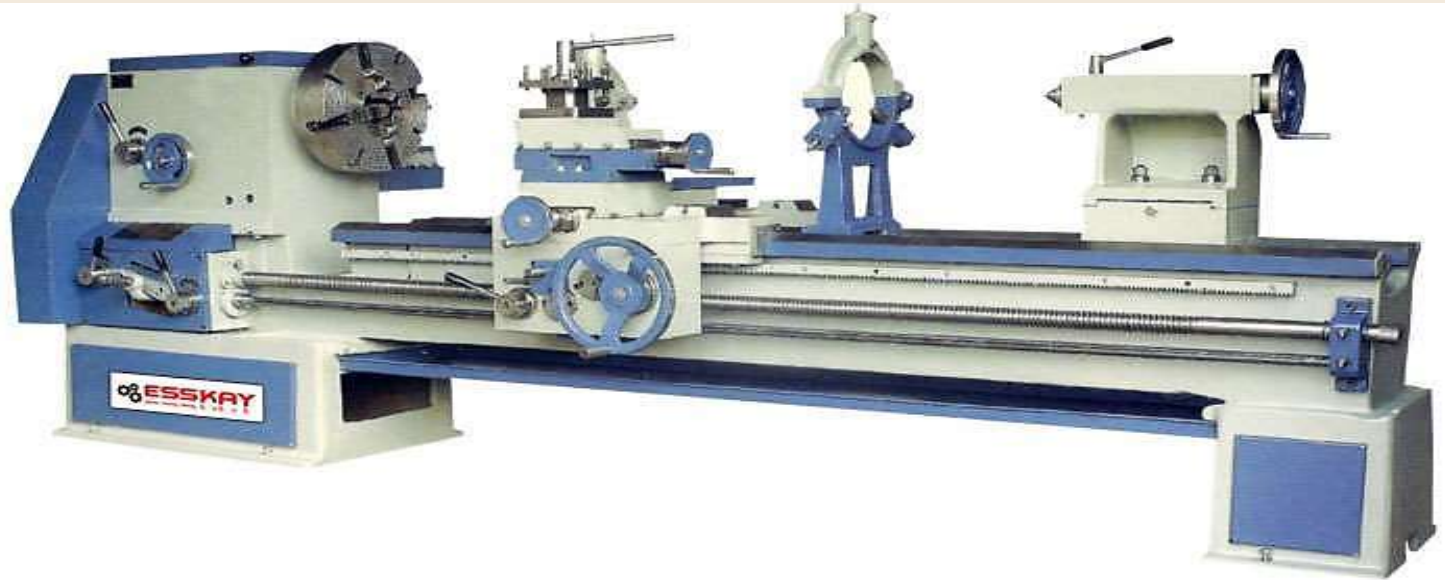


Size of Lathe machine

Example: 300 - 1500 Lathe

- Maximum Diameter of Workpiece that can be machined
= **SWING** (= 300 mm)
- Maximum Length of Workpiece that can be held between Centers (=1500 mm)

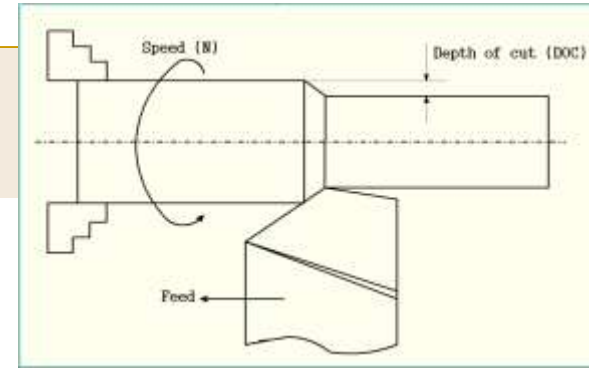
Specification of Lathe Machine



SPECIFICATION OF LATHE MACHINE

MODEL	EKL - 614	EKL - 816	EKL - 1018	EKL - 1220	EKL - 1222	EKL- 1224
Length of bed	1828 MM	2438 MM	3048 MM	3657 MM	3657 MM	3657 MM
Width of bed	350 MM	400 MM	450 MM	500 MM	550 MM	600 MM
Height of Center	300 MM	350 MM	400 MM	450 MM	500 MM	550 MM
Swing Over Slide	430 MM	533 MM	635 MM	736 MM	787 MM	863 MM
Swing in Gap	914 MM	1016 MM	1117 MM	1370 MM	1422 MM	1473 MM
Admit Between Centers	838 MM	1371 MM	1980 MM	2413 MM	2413 MM	2363 MM
Spindle Hollow	53 MM	63 MM	88 MM	103 MM	103 MM	103 MM
Type of Bed	Pedestal	Pedestal	Pedestal	Planner	Planner	Planner
Hp. Required	2*960	2*960	3*960	3*960	3*960	5*960
T.P.L of Lead Screw	4	2	2	2	2	2

Cutting parameters



□ Speed

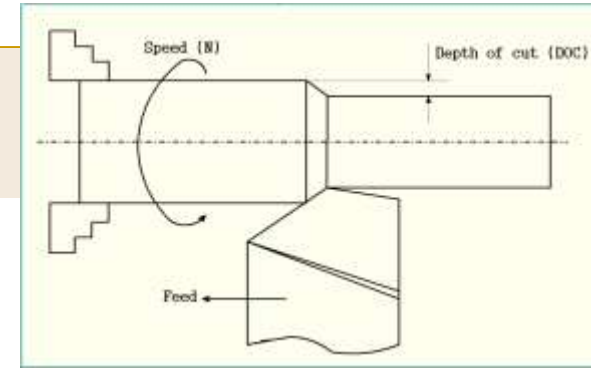
Cutting speed is defined as the distance at which the work moves with respect to the tool (usually measured in mm/minute).

□ Feed

Feed rate is defined as the distance the tool travels during one revolution of the part.

□ Depth of Cut (DOC) is the perpendicular thickness of material removed from a workpiece in a single pass, directly affecting cutting forces, tool life, and surface finish

Cutting parameters

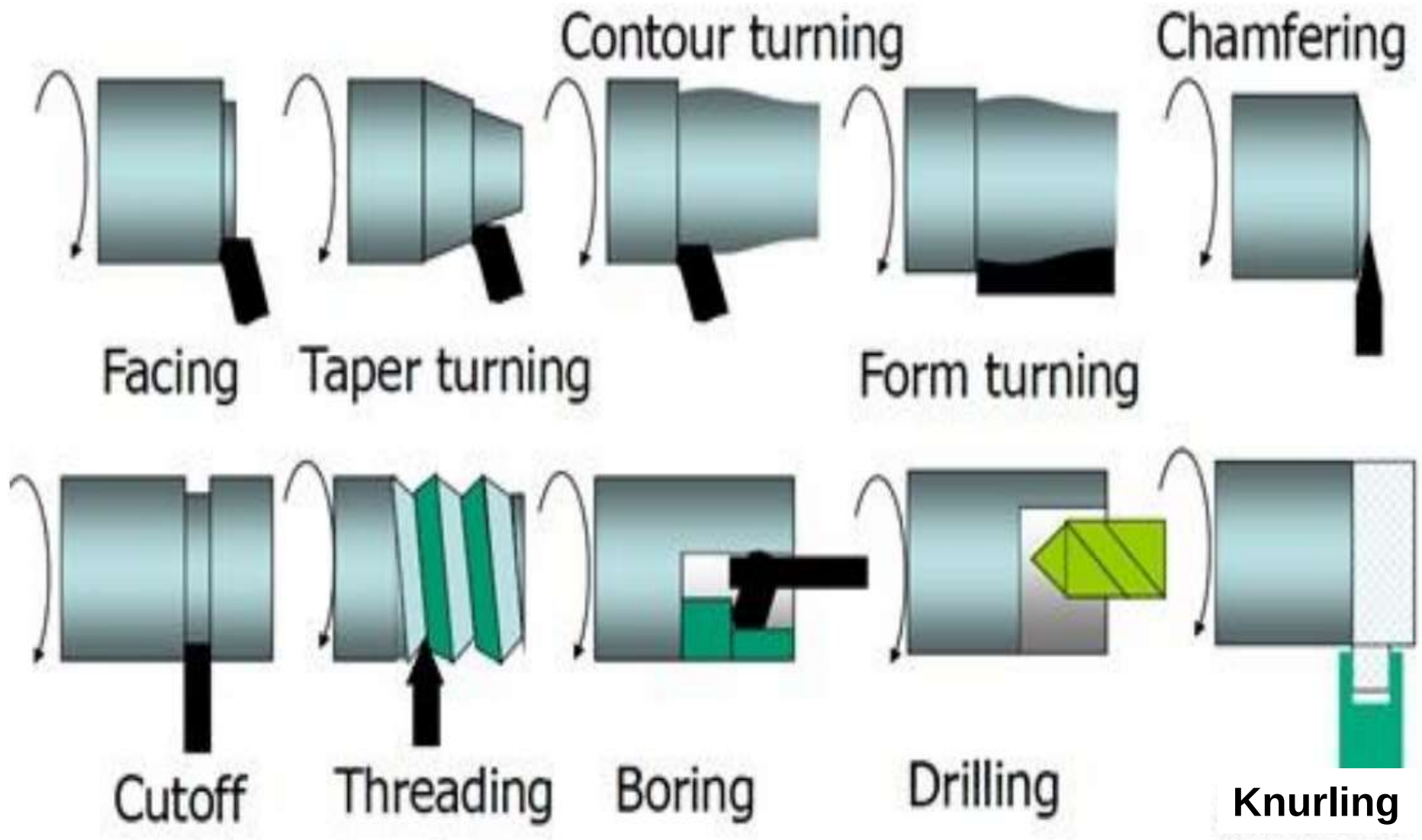


- Cutting speed and feed determines the surface finish, power requirements, and material removal rate.
- The primary factor in choosing **feed and speed** is the material to be cut.
- However, one should also consider **material of the tool**, rigidity of the work piece, size and condition of the lathe, and depth of cut.

Key Components of Taylor's Tool Life Equation

- Equation: $VT^n = C$
- **V**(Cutting Speed): Velocity in *m/min*
- **T**(Tool Life): Time to failure in minutes
- **n** (Tool Life Exponent): Varies by tool material (e.g., 0.08 – 0.02 for HSS, 0.2 – 0.4 for Carbide)
- **C** (Machining Constant): Depends on tool-work material and cutting Environment .

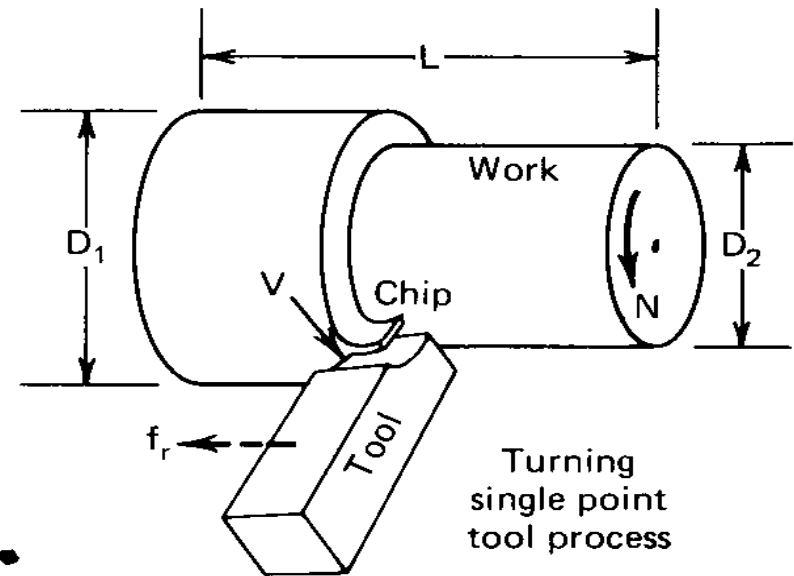
Lathe operations



Machining operations

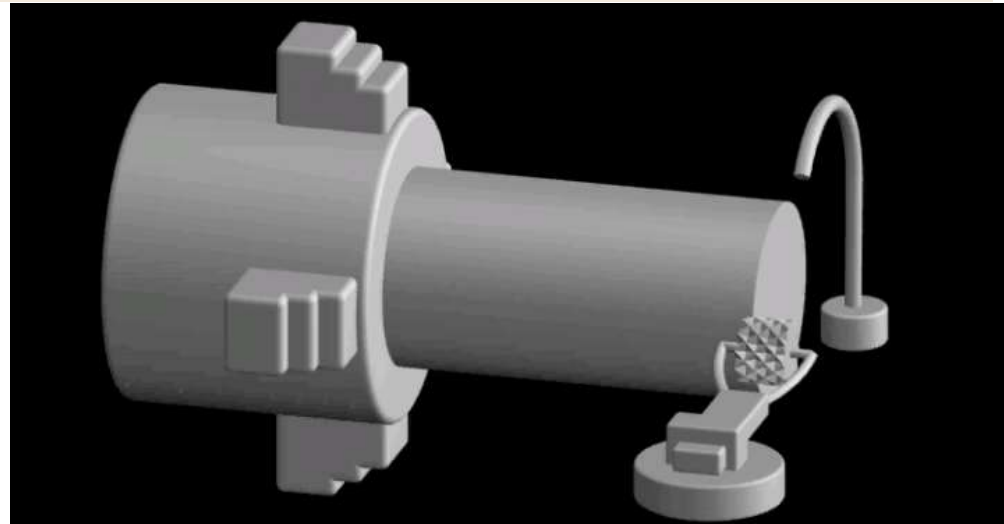
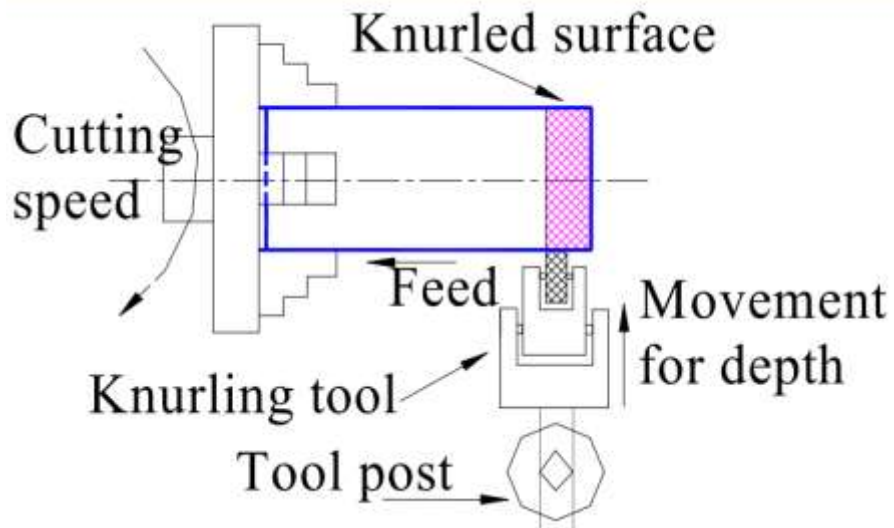
- There are many kinds of machining operations, each of which is capable of generating a certain part geometry and surface texture.
- In **turning**, a cutting tool with a single cutting edge is used to remove material from a rotating workpiece to generate a cylindrical shape. The primary motion is provided by rotating the workpiece, and the feed motion is achieved by moving the cutting tool slowly in a direction parallel to the axis of rotation of the workpiece.
- **Drilling** is used to create a round hole. It is accomplished by a rotating tool that typically has two or four helical cutting edges. The tool is fed in a direction parallel to its axis of rotation into the workpiece to form the round hole.
- **Shaping** process in which cutting motion provided by linear-forward motion of the reciprocating tool and the intermittent feed motion provided by the slow transverse motion of the workpiece resulting in a flat surface by gradual removal of excess material layer by layer.
- In **boring**, a tool with a single bent pointed tip is advanced into a roughly made hole in a spinning workpiece to slightly enlarge the hole and improve its accuracy. It is a fine finishing operation used in the final stages of product manufacture.
- In **milling**, a rotating tool with multiple cutting edges is moved slowly relative to the material to generate a plane or straight surface. The direction of the feed motion is perpendicular to the tool's axis of rotation. The speed motion is provided by the rotating milling cutter.

Turning Operation



- Machine Tool – LATHE
- Job (workpiece) – rotary motion
- Tool – linear motions
 - ❖ "Mother of Machine Tools"
 - ❖ Cylindrical and flat surfaces

Machine tools

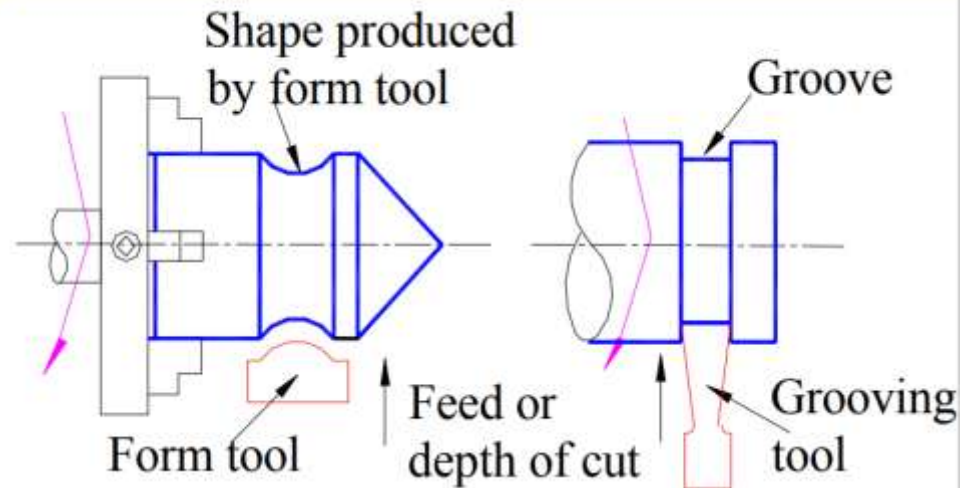


Knurling

It is used to produce regular patterned rough surface. Knurling tool containing a set of hardened steel rollers with teeth cut on them is used. The metal is squeezed against the multiple edges. The speed should be low and plenty of lubricant should be used.

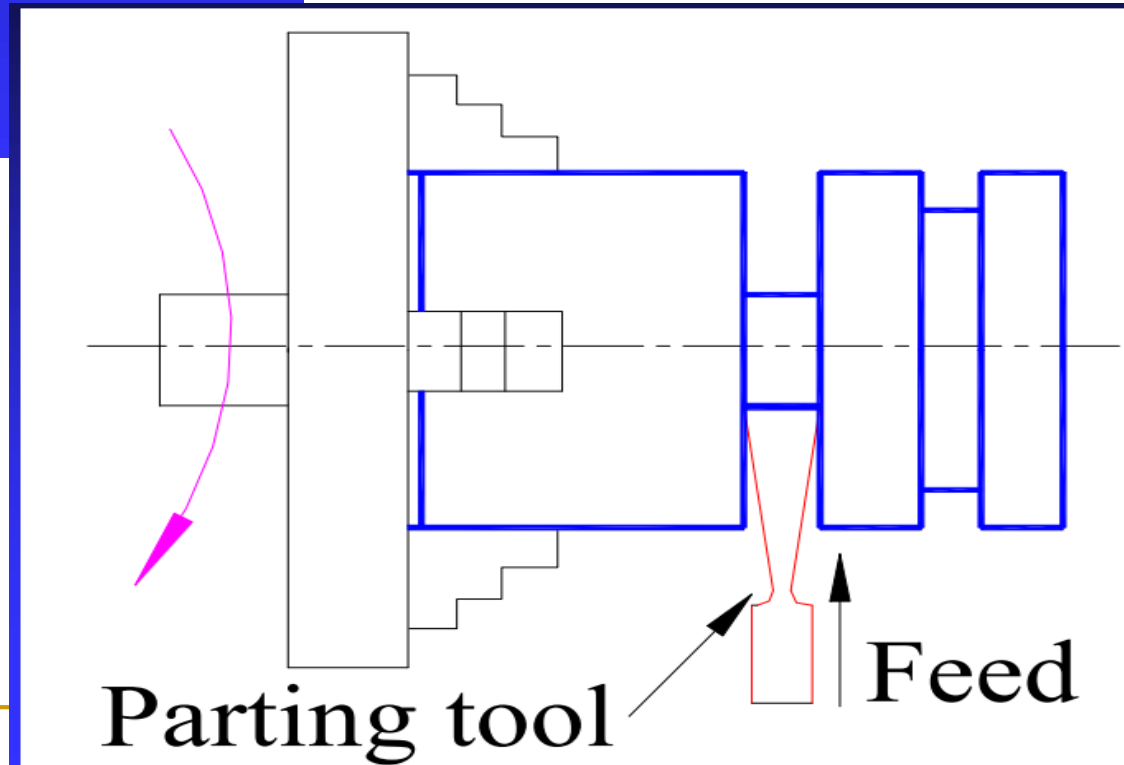
Grooving Operation

- Produces a **Groove** on workpiece
- Shape of tool $\Rightarrow$ shape of groove
- Carried out using **Grooving Tool** $\Rightarrow$ A **form tool**
- Also called **Form Turning**



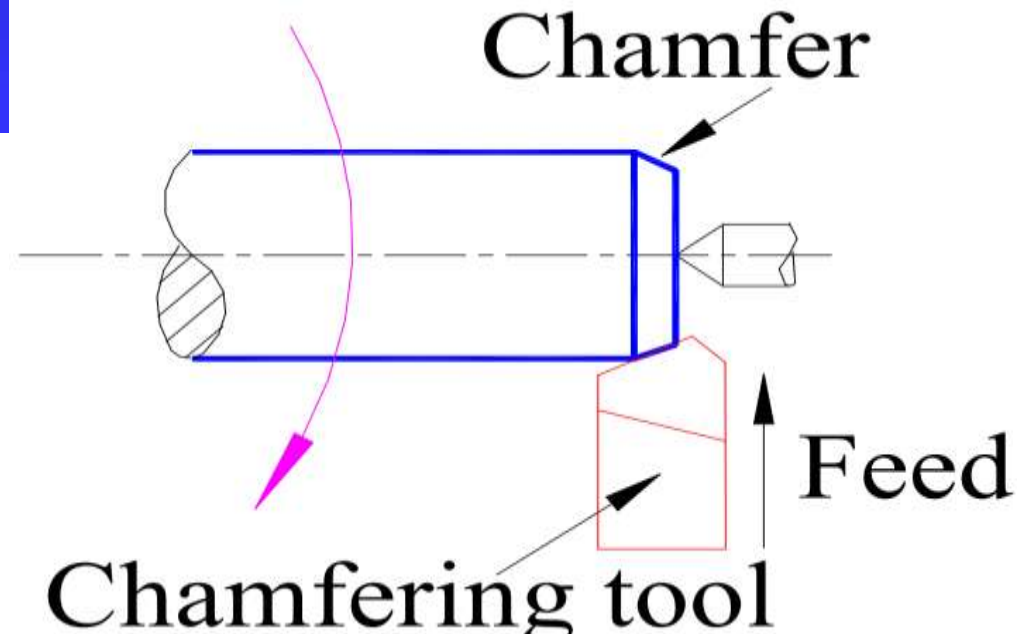
Parting

- Cutting workpiece into **Two**
- Similar to grooving
- **Parting Tool**
- *Hogging* – tool rides over – at slow feed
- Coolant use



Chamfering

- ✧ Beveling sharp machined edges
- ✧ Similar to form turning
- ✧ Chamfering tool – 45°
- ✧ To
 - ◆ Avoid Sharp Edges
 - ◆ Make Assembly Easier
 - ◆ Improve Aesthetics



Taper Turning

Conicity

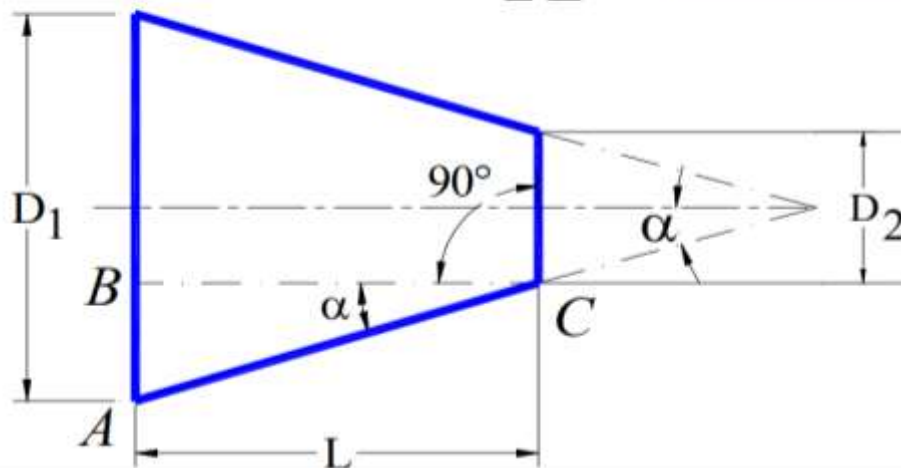
$$K = \frac{D_1 - D_2}{L}$$

Methods

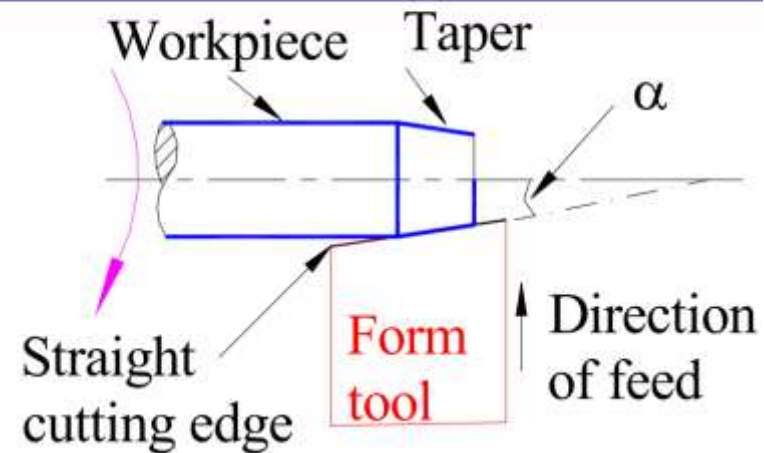
- Form Tool
- Swiveling Compound Rest
- Taper Turning Attachment
- Simultaneous Longitudinal and Cross Feeds

■ Taper:

$$\tan \alpha = \frac{D_1 - D_2}{2L}$$



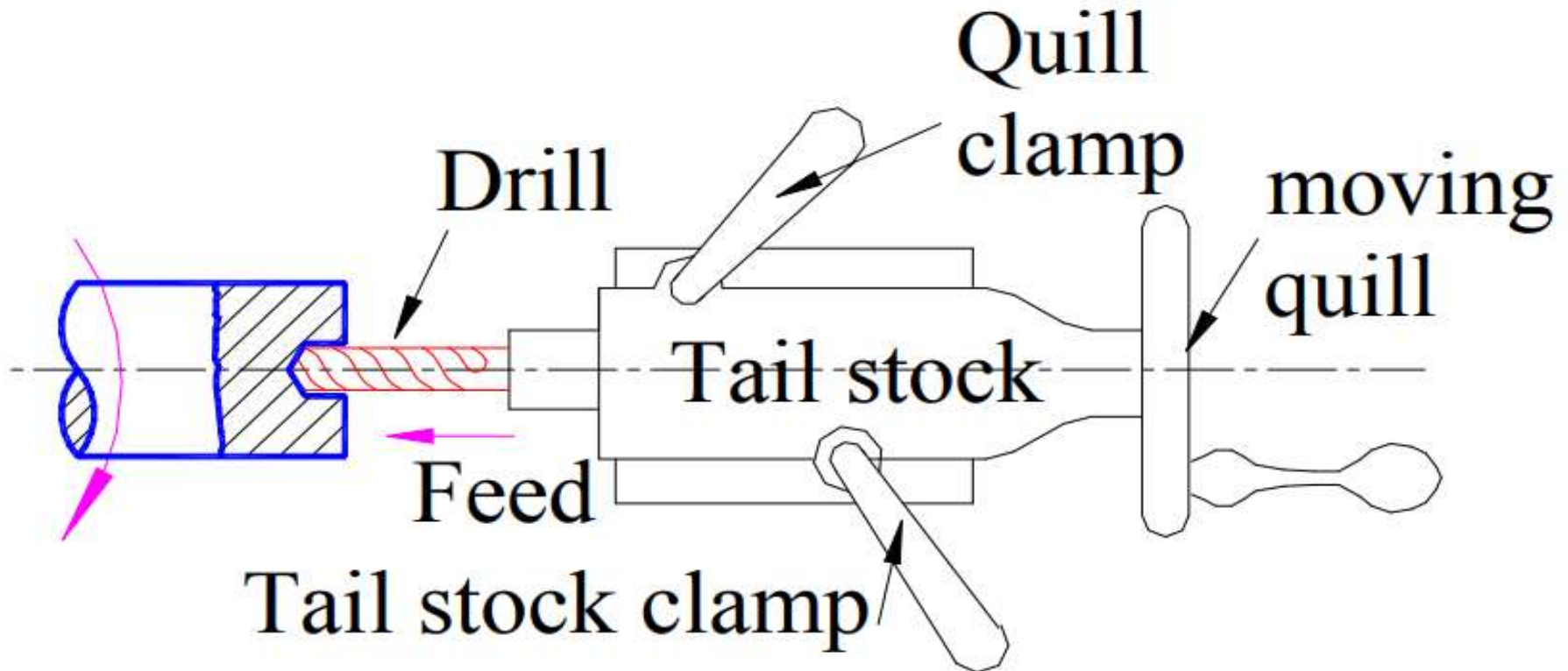
Taper Turning .. By Form Tool



Drilling

Drill – cutting tool – held in TS –
feed from TS

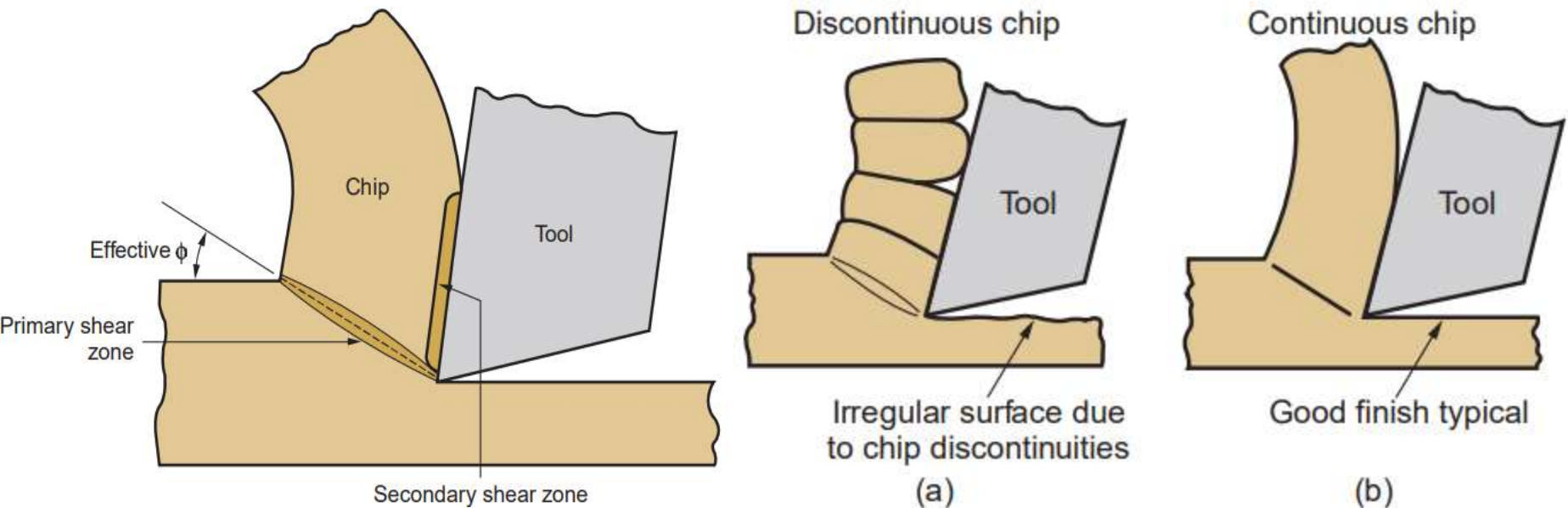
TS - Tailstock



Actual Chip Formation

Discontinuous chip. When relatively brittle materials (e.g., cast irons) are machined at low cutting speeds, the chips often form into separate segments (sometimes the segments are loosely attached).

Continuous chip. When ductile work materials are cut at high speeds and relatively small feeds and depths, long continuous chips are formed. A good surface finish typically results when this chip type is formed.



Forces in Metal Cutting

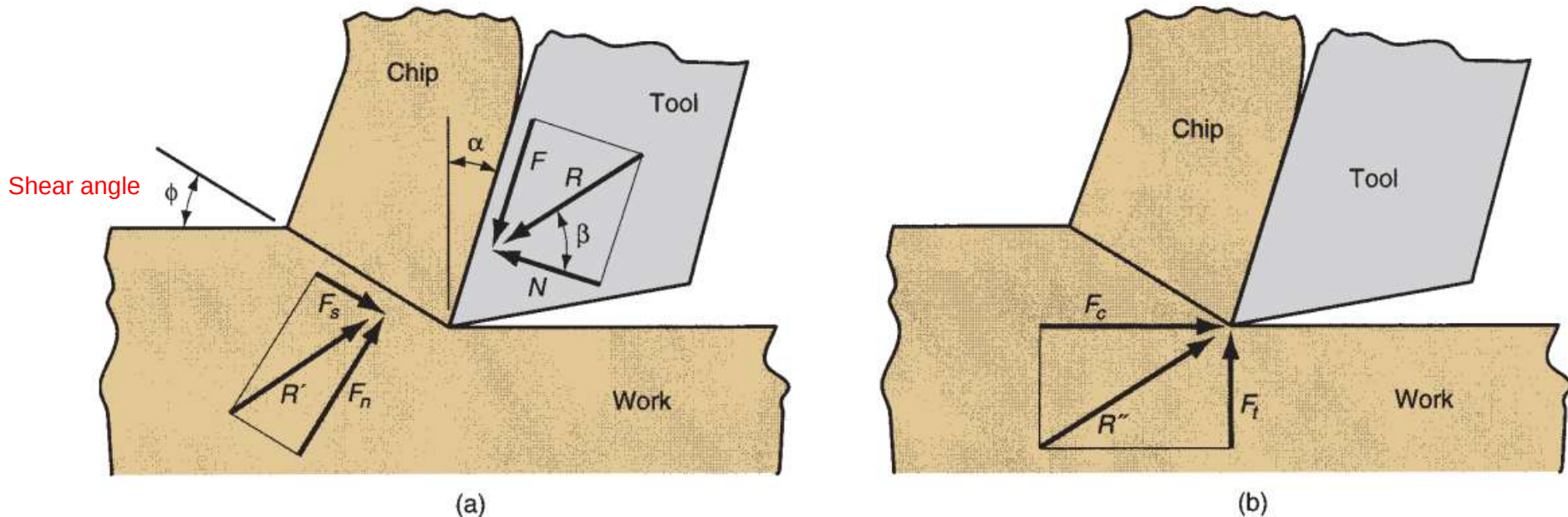


FIGURE 20.10 Forces in metal cutting: (a) forces acting on the chip in orthogonal cutting, and (b) forces acting on the tool that can be measured.

$$\mu = \frac{F}{N} \quad \tau = \frac{F_s}{A_s} \quad (20.7)$$

where A_s = area of the shear plane. This shear plane area can be calculated as

$$\mu = \tan \beta \quad A_s = \frac{t_o w}{\sin \phi} \quad (20.8)$$

Forces in Metal Cutting

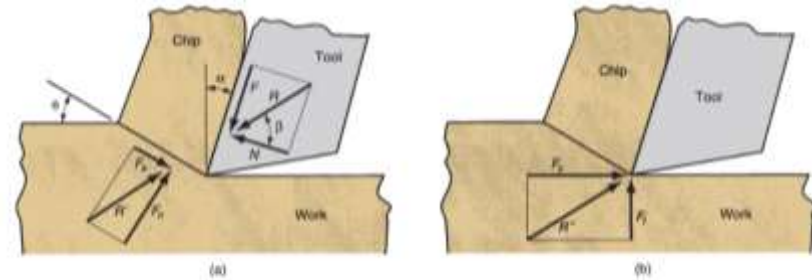
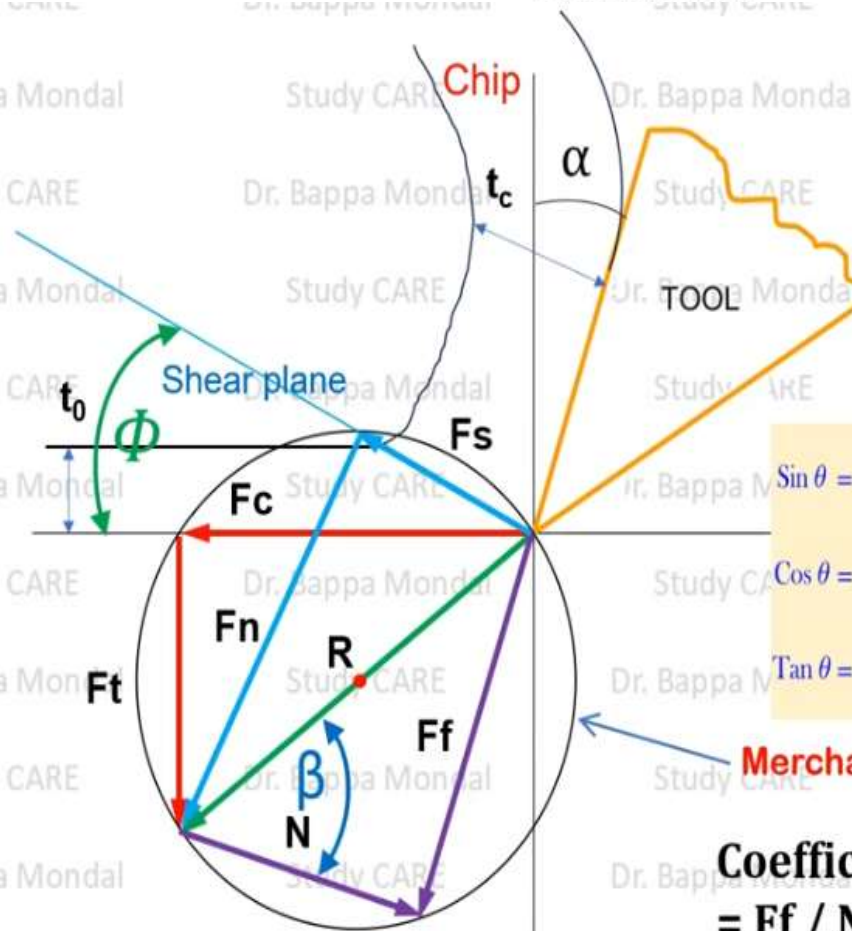


FIGURE 20.10 Forces in metal cutting: (a) forces acting on the chip in orthogonal cutting, and (b) forces acting on the tool that can be measured.

Merchant circle diagram (MCD)

α = rake angle
 β = Friction angle
 Φ = Shear angle

F_c Cutting Force
 F_t Tangential Force
 F_f Friction Force
 N Normal Force
 F_s Shear Force
 F_n Normal Shear Force



$$\begin{aligned} \sin \theta &= \frac{P}{H} & \operatorname{cosec} \theta &= \frac{H}{P} \\ \cos \theta &= \frac{B}{H} & \sec \theta &= \frac{H}{B} \\ \tan \theta &= \frac{P}{B} & \cot \theta &= \frac{B}{P} \end{aligned}$$

P- Perpendicular
 B- Base
 H- Hypotenuse

Merchant circle

Coefficient of friction
 $= F_f / N$

<https://www.youtube.com/watch?v=QmdzG6ytizI>

Forces in Metal Cutting

Equations can be derived to relate the four force components that cannot be measured to the two forces that can be measured. Using the force diagram in Figure 20.11, the following trigonometric relationships can be derived:

$$F = F_c \sin \alpha + F_t \cos \alpha \quad (20.9)$$

$$N = F_c \cos \alpha - F_t \sin \alpha \quad (20.10)$$

$$F_s = F_c \cos \phi - F_t \sin \phi \quad (20.11)$$

$$F_n = F_c \sin \phi + F_t \cos \phi \quad (20.12)$$

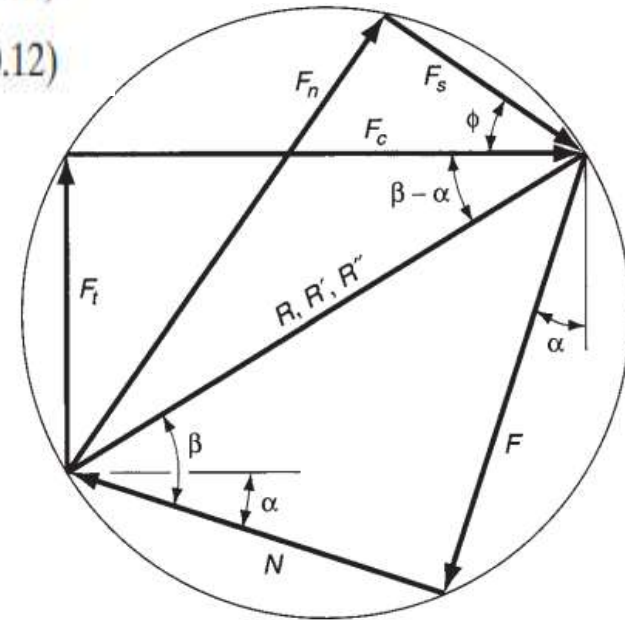
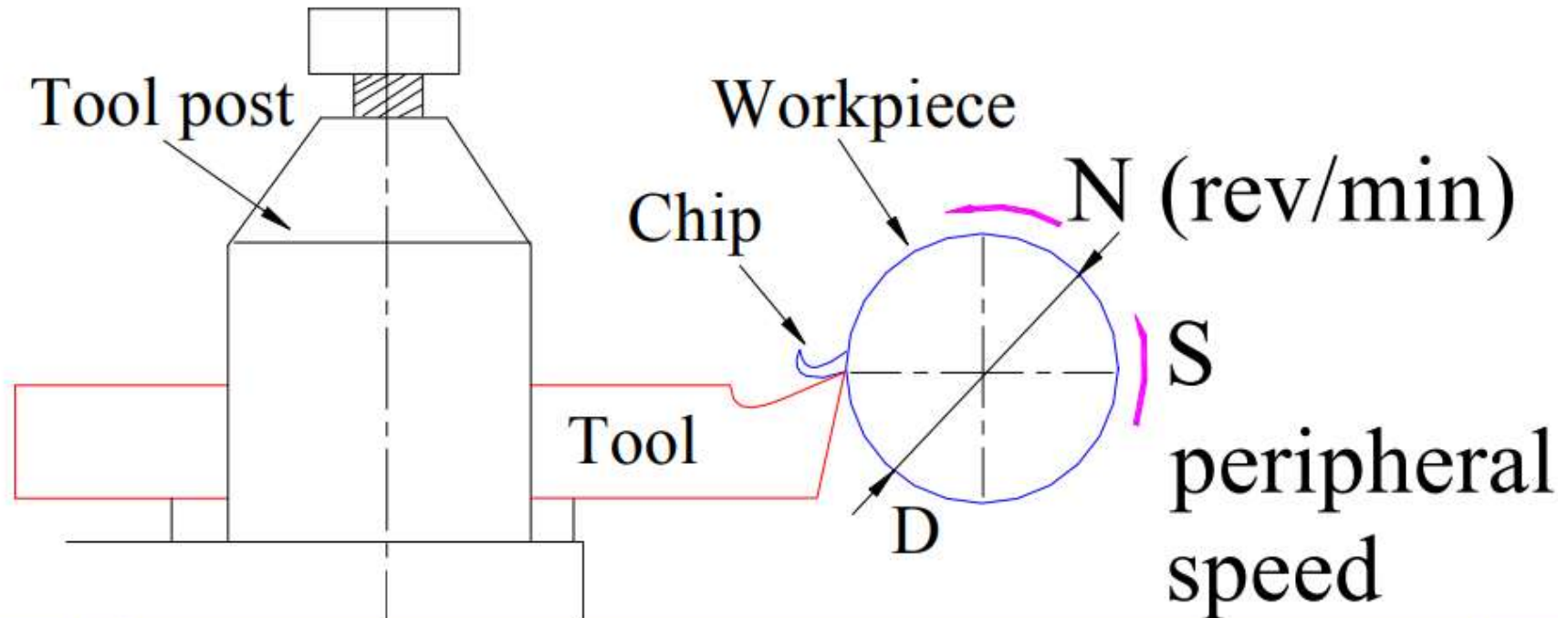


FIGURE 20.11 Force diagram showing geometric relationships between F , N , F_s , F_n , F_c , and F_t .

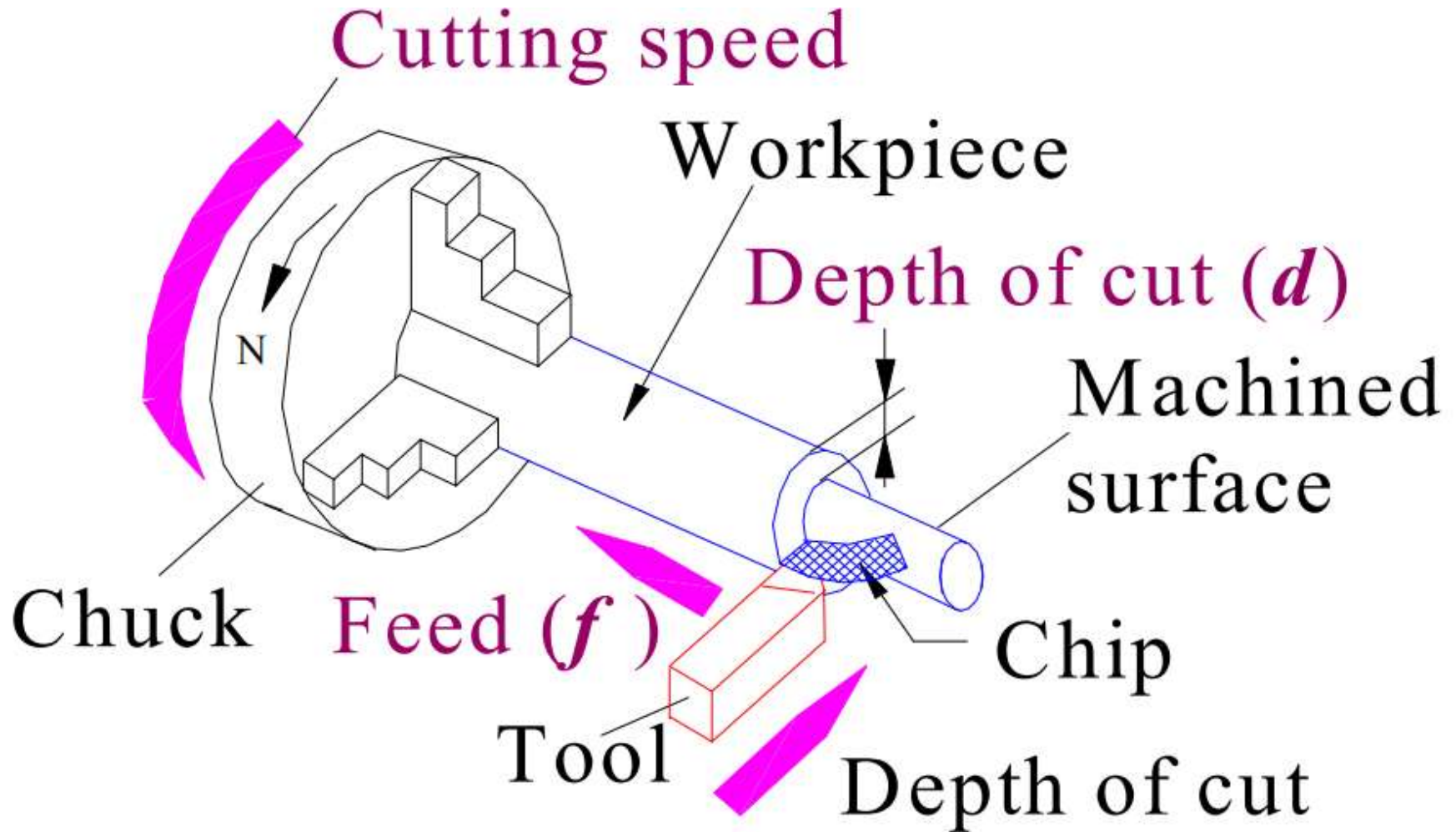
Operating Conditions



relative tool travel in 1 rotation = πD

$$\text{peripheral speed} = S = \pi D N$$

Operating Conditions...



Cutting Speed

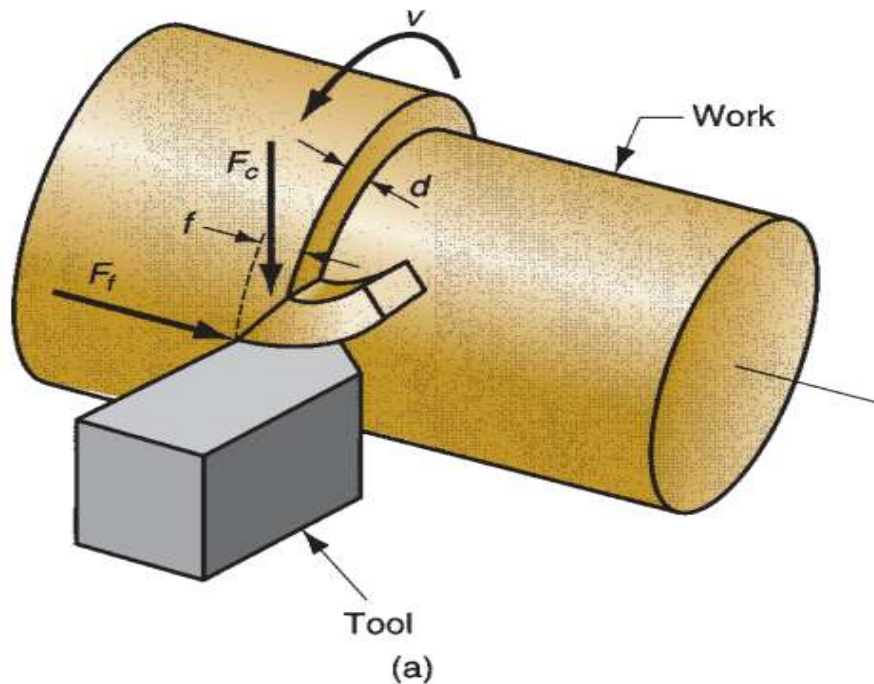
D – Diameter (mm)

N – Revolutions per Minute (rpm)

$$v = \frac{\pi D N}{1000} \quad \text{m/min}$$

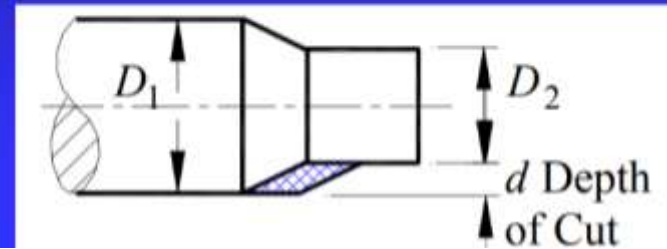
The **Peripheral Speed** of
Workpiece past the Cutting Tool
= Cutting Speed

Feed and Depth of Cut



perpendicular distance between machined surface and uncut surface of the Workpiece

$$d = (D_1 - D_2)/2 \quad (\text{mm})$$



Material Removal Rate (MRR)

MRR

Volume of material removed in one revolution $MRR = \pi D d f \text{ mm}^3$

- Job makes N revolutions/min

$$MRR = \pi D d f N \text{ (mm}^3\text{/min)}$$

- In terms of v MRR is given by

$$MRR = 1000 v d f \text{ (mm}^3\text{/min)}$$

dimensional consistency by
substituting the units

$$\begin{aligned} MRR: D d f N &\Rightarrow \\ (\text{mm})(\text{mm})(\text{mm/rev})(\text{rev/min}) & \\ = \text{mm}^3/\text{min} & \end{aligned}$$

Machining Time for Turning operation

- Job length L_j mm
- Feed f mm/rev
- Job speed N rpm
- $f N$ mm/min

$$t = \frac{L_j}{f N} \text{ min}$$

002

Manufacturing Time
= Machining Time
+ Setup Time
+ Moving Time
+ Waiting Time

Notations

N = Rotational speed of the workpiece, rpm

f = Feed, mm/rev or in./rev

v = Feed rate, or linear speed of the tool along workpiece length, mm/min or in./min
 $= fN$

V = Surface speed of workpiece, m/min or ft/min
 $= \pi D_o N$ (for maximum speed)
 $= \pi D_{avg} N$ (for average speed)

l = Length of cut, mm or in.

D_o = Original diameter of workpiece, mm or in.

D_f = Final diameter of workpiece, mm or in.

D_{avg} = Average diameter of workpiece, mm or in.
 $= (D_o + D_f)/2$

d = Depth of cut, mm or in.
 $= (D_o + D_f)/2$

t = Cutting time, s or min
 $= l/fN$

MRR = mm³/min or in.³/min
 $= \pi D_{avg} d fN$

Numerical 1

1. A cylindrical stainless steel rod with length $L=150$ mm, diameter $D_o = 12$ mm is being reduced in diameter to $D_f=11$ mm by turning on a lathe. The spindle rotates at $N = 400$ rpm, and the tool is travelling at an axial speed of $v=200$ mm/min

Calculate:

- a. The cutting speed V (maximum and minimum)
- b. The material removal rate MRR
- c. The cutting time t
- d. The power required if the unit power is estimated to 4 w.s/mm³

Solution

a. The maximum cutting speed is at the outer diameter D_o and is obtained from the expression $V = \pi D_o N$

Thus,

$$V_{\max} = (\pi) (12) (400) = 15072 \text{ mm/min}$$

The cutting speed at the inner diameter D_f is

$$V_{\min} = (\pi) (11) (400) = 13816 \text{ mm/min}$$

b. From the information given, the depth of cut is

$$d = (12 - 11) / 2 = 0.5 \text{ mm}, \text{ and the feed is } f = v / N$$

$$f = 200 / 400 = 0.5 \text{ mm/rev}$$

thus the material removal rate is calculated as

$$\text{MRR} = (\pi) (D_{\text{avg}}) (d) (f) (N) = (\pi) (11.5) (0.5) (0.5) (400) = 3611 \text{ mm}^3 / \text{min} = 60.2 \text{ mm}^3 / \text{s}$$

c. The cutting time is

$$t = l / (f \cdot N) = (150) / (0.5) (400) = 0.75 \text{ min}$$

d. The power required is

$$\text{Power} = (4) (60.2) = 240.8 \text{ W}$$

Problem 2

A batch of 800 workpieces is to be produced on a turning machine. Each workpiece with length $L=120\text{mm}$ and diameter $D=10\text{mm}$ is to be machined from a raw material of $L=120\text{mm}$ and $D=12\text{mm}$ using a cutting speed $V=32\text{ m/min}$ and a feed rate $f=0.8\text{ mm/rev}$. How many times we have to resharpen or regrind the cutting tool?

In the Taylor's expression, use constants as $n=1.25$ and $C=175$.

Solution for prob. 2

Solution. Given : $L = 120$ mm, $D_i = 12$ mm ; $D_f = 10$ mm ; $V = 32$ m/min ; $f = 0.8$ mm rev ;
 $n = 1.25$; $C = 175$.

No. of resharpenings required :

From Taylor's tool life equation, we have

$$VT^n = C$$

$$32 \times (T)^{1.25} = 175$$

or,
$$T = \left(\frac{175}{32} \right)^{1/1.25} = 3.89 \text{ min}$$

Also,
$$V = \frac{\pi DN}{1000}$$

or,
$$32 = \frac{\pi \times 12 \times N}{1000}$$

or,
$$N \simeq 849 \text{ r.p.m.}$$

Now, **Machining time**/piece = $\frac{L}{f \times N} = \frac{120}{0.8 \times 849} = 0.177 \text{ min.}$

$\therefore$ **Machining time** for 800 workpieces = $800 \times 0.177 = 141.6 \text{ min.}$

Hence, *number of resharpenings required* = $\frac{141.6}{3.89} = 36.4$ or **36 resharpenings (Ans.)**

Problem 3

A mild steel rod having 50 mm diameter and 500 mm length is to be turned on a lathe.

Determine the machining time to reduce the rod to 45 mm in one pass when cutting speed is 30 m/min and a feed of 0.7 mm/rev is used.

Solution for prob. 3

Given data: $D = 50$ mm, $L_j = 500$ mm
 $v = 30$ m/min, $f = 0.7$ mm/rev

Substituting the values of v and D in

$$v = \frac{\pi D N}{1000} \text{ m/min}$$

calculate the required spindle speed as: $N = 191$ rpm

Can a machine has speed of 191 rpm?

Machining time:

$$t = \frac{L_j}{f N} \text{ min}$$

$$\begin{aligned} \blacksquare t &= 500 / (0.7 \times 191) \\ \blacksquare &= 3.74 \text{ minutes} \end{aligned}$$

Machinability

- The major factor involved in metal cutting are,
 - Forces and power absorbed
 - Tool wear and tool life
 - Surface finish
 - Dimensional accuracy
 - Machining cost
- This factor depend upon a large variables, such as
 - Property of work material
 - Tool geometry
 - Cutting condition
 - Machine tool rigidity

Qualitative measure of machinability

1. The easy with which it could be machined,
2. The life of tool before tool failure or re sharpening
3. The quality of machined surface.
4. The power consumption per unit volume of material removed.

The machinability may be evaluated as:

- Long tool life at a given cutting speed
- Low power consumption per unit volume of material removed.
- Maximum metal removal per tool re sharpening
- High quality of surface finish
- Good and uniform dimensional accuracy of successive parts
- Easy disposable chips.

What is Cutting Fluid ?

- **Cutting fluid** is a type of coolant and lubricant designed specifically for metalworking and machining processes.
- There are various kinds of cutting fluids, which include oils, oil-water emulsions, pastes, gels and other gases.
- They may be made from petroleum distillates, animal fats, plant oils, water and other raw ingredients.
- Depending on context, type of cutting fluid is being considered, it may be referred to as **cutting fluid**, **cutting oil**, **cutting compound**, **coolant**, or **lubricant**.



Economic Advantages to Using Cutting Fluids

- **Reduction of tool costs**
 - Reduce tool wear, tools life longer
 - **Increased speed of production**
 - Reduce heat and friction so higher cutting speeds
 - **Reduction of labor costs**
 - Tools life longer and require less regrinding, less downtime, reducing cost per part
 - **Reduction of power costs**
 - Friction reduced so less power required by machining
-

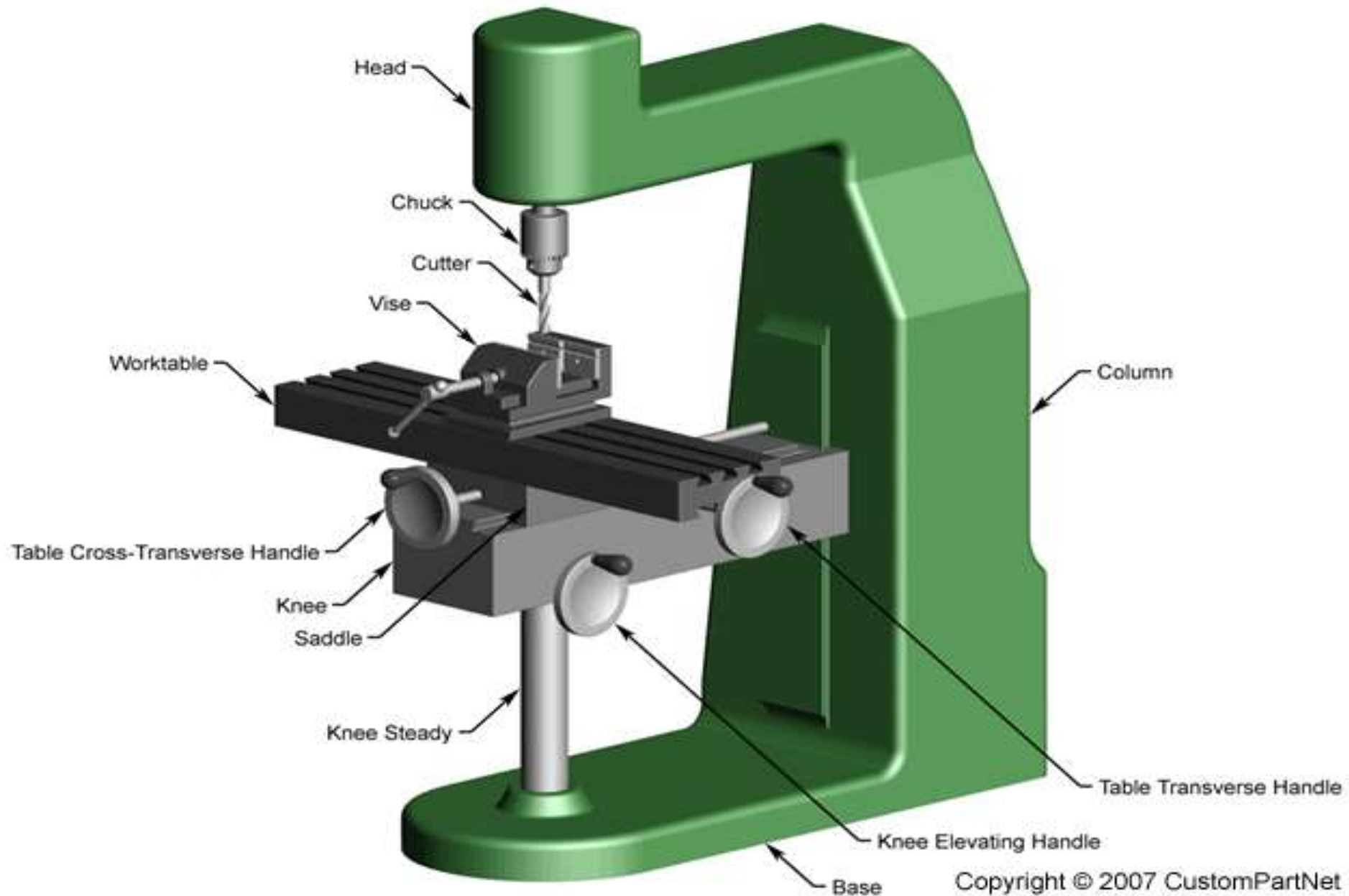
Characteristics of a Good Cutting Fluid

- Good cooling capacity
- Good lubricating qualities
- Relatively low viscosity
- Stability (long life)
- Rust resistance
- Nontoxic
- Transparent
- Nonflammable

Types of Cutting Fluids

- Most commonly used cutting fluids
 - Either aqueous based solutions or cutting oils
- Three categories
 - Cutting oils
 - Emulsifiable oils
 - Chemical (synthetic) cutting fluids

Milling Machine tool



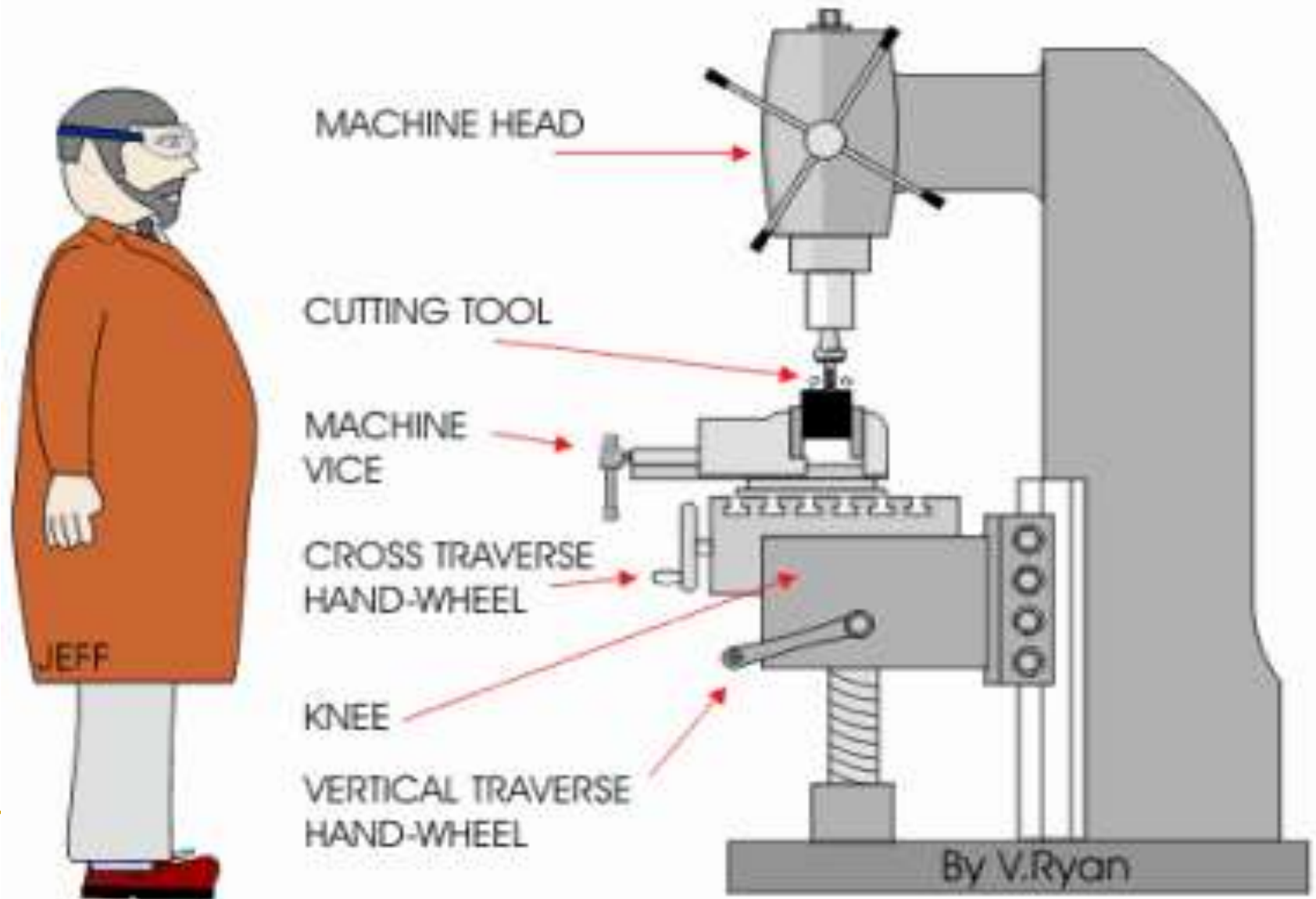
Milling Machine Tool

- Milling is one of the basic machining processes.
- Milling is a very versatile process capable of producing simple two dimensional flat shapes to complex three dimensional interlaced surface configurations.

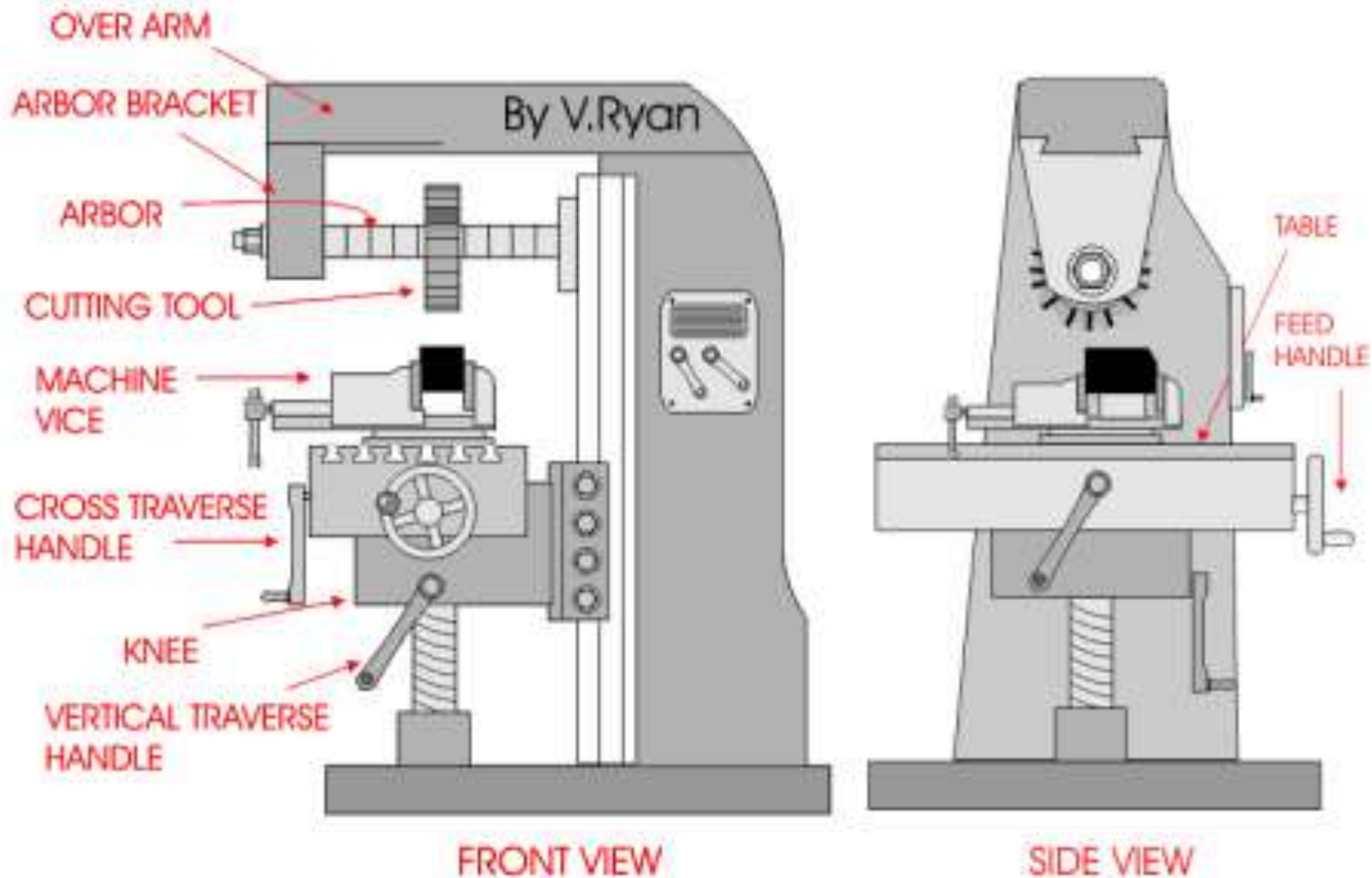
■ The milling process:

- Typically uses a multi-tooth cutter
- Work is fed into the rotating cutter
- Capable of high MRR
- Well suited for mass production applications
- Cutting tools for this process are called milling cutters

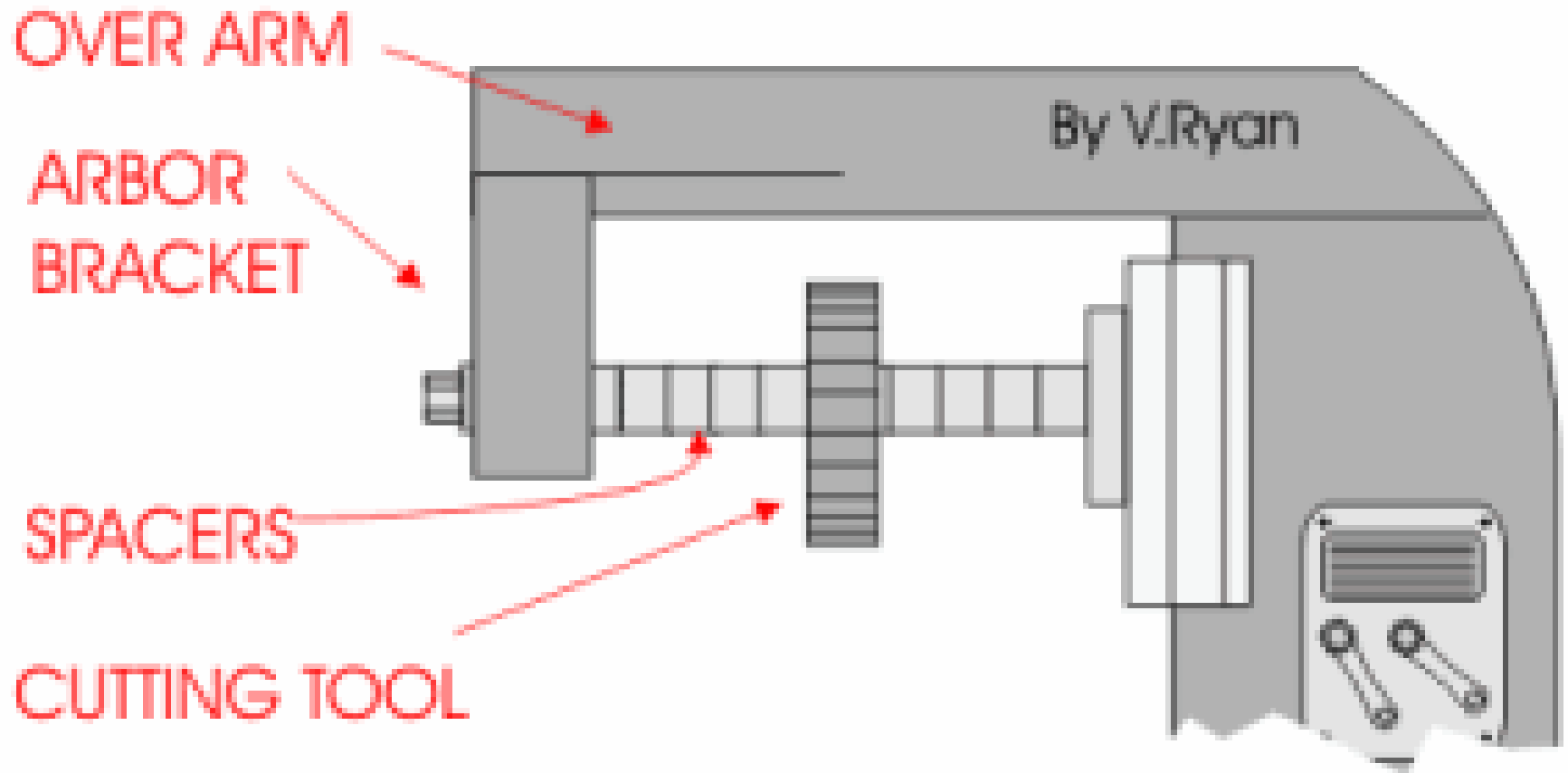
Vertical Mill



Horizontal Mill



Horizontal Mill

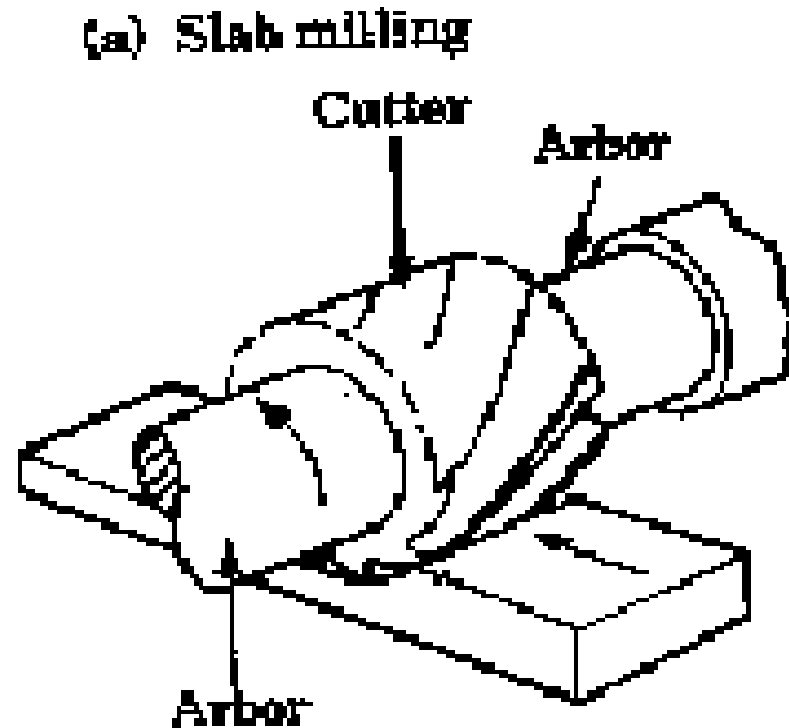


Classification of Milling

- Peripheral Milling (SLAB Milling)
- Face Milling
- End Milling

Peripheral Milling (SLAB Milling)

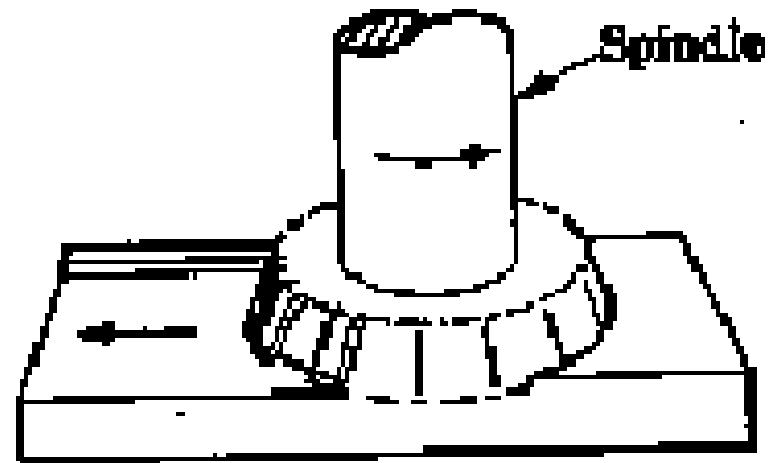
- The milled surface is generated by **teeth located on the periphery of the cutter body**.
- The axis of cutter rotation is generally in a plane **parallel** to the workpiece surface to be machined.



Face Milling

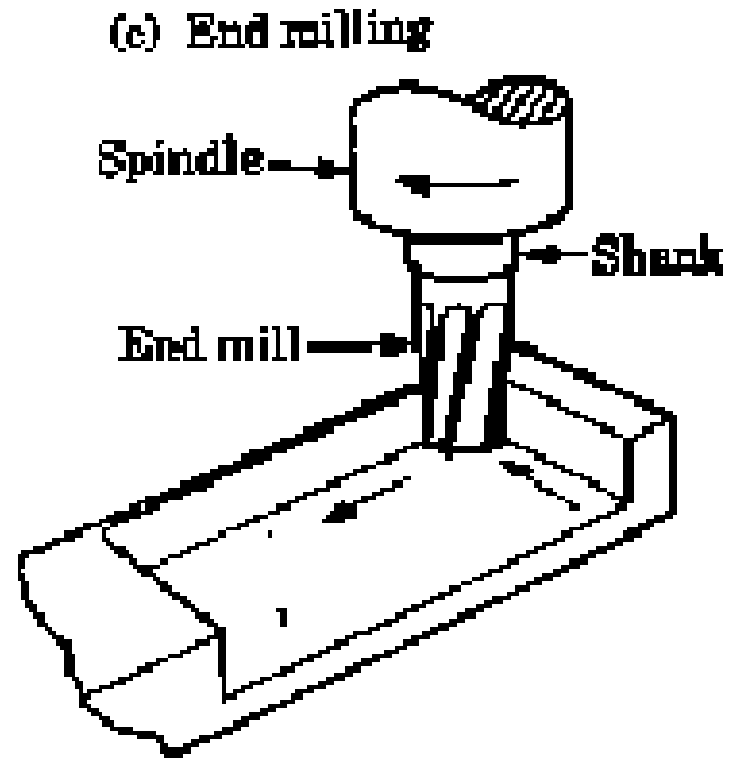
- The cutter is mounted on a spindle having an axis of rotation **perpendicular** to the workpiece surface.
- The milled surface results from the action of **cutting edges located on the periphery and face of the cutter**

(b) Face milling



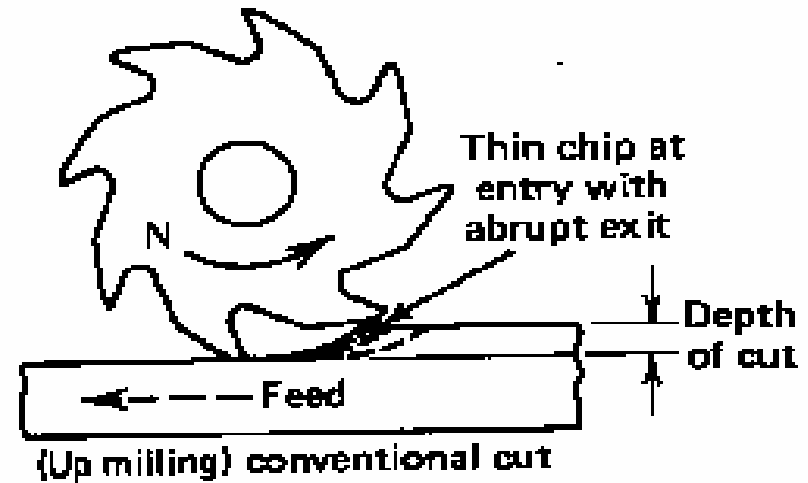
End Milling

- The cutter in end milling generally rotates on an axis vertical to the workpiece.
- It can be tilted to machine tapered surfaces.
- Cutting teeth are located on both the end face of the cutter and the periphery of the cutter body.



Methods of milling

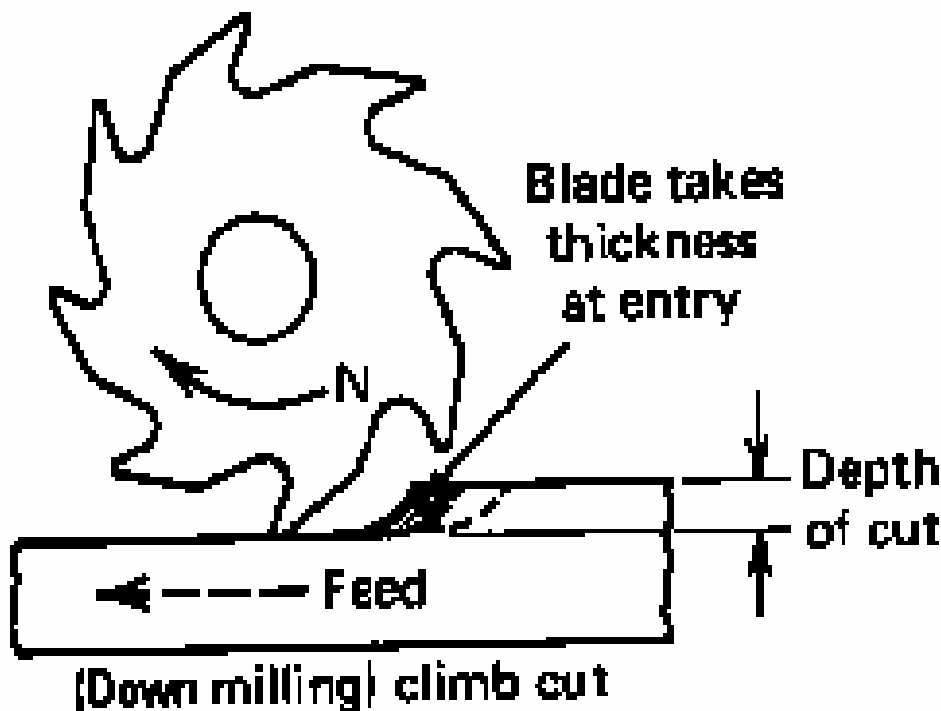
Up Milling



Peripheral or
slab milling

- The direction of cutter rotation is same as the feed motion.
- For example, if the cutter rotates clockwise, the work-piece is fed to the right in down milling.

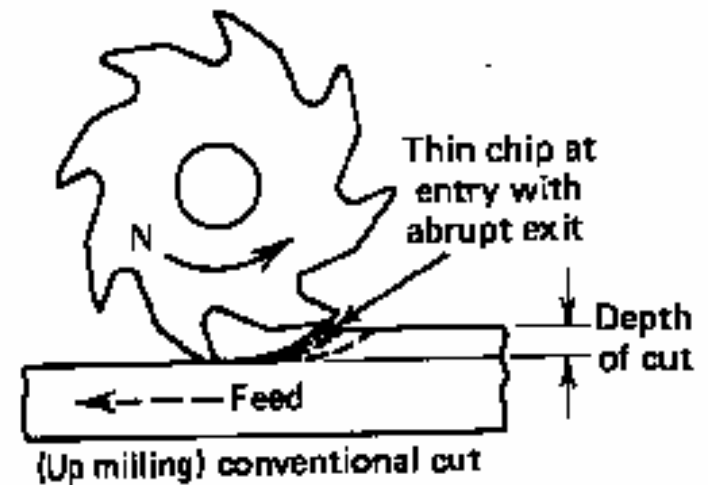
Down Milling



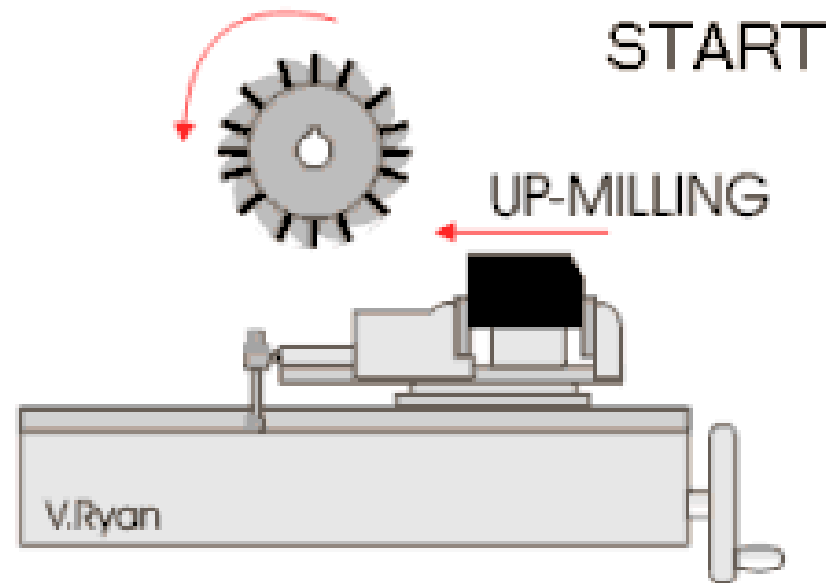
UP MILLING :

(conventional milling)

- The direction of the cutter rotation opposes the feed motion.
- For example, if the cutter rotates counterclockwise, the work-piece is fed to the right in up milling.



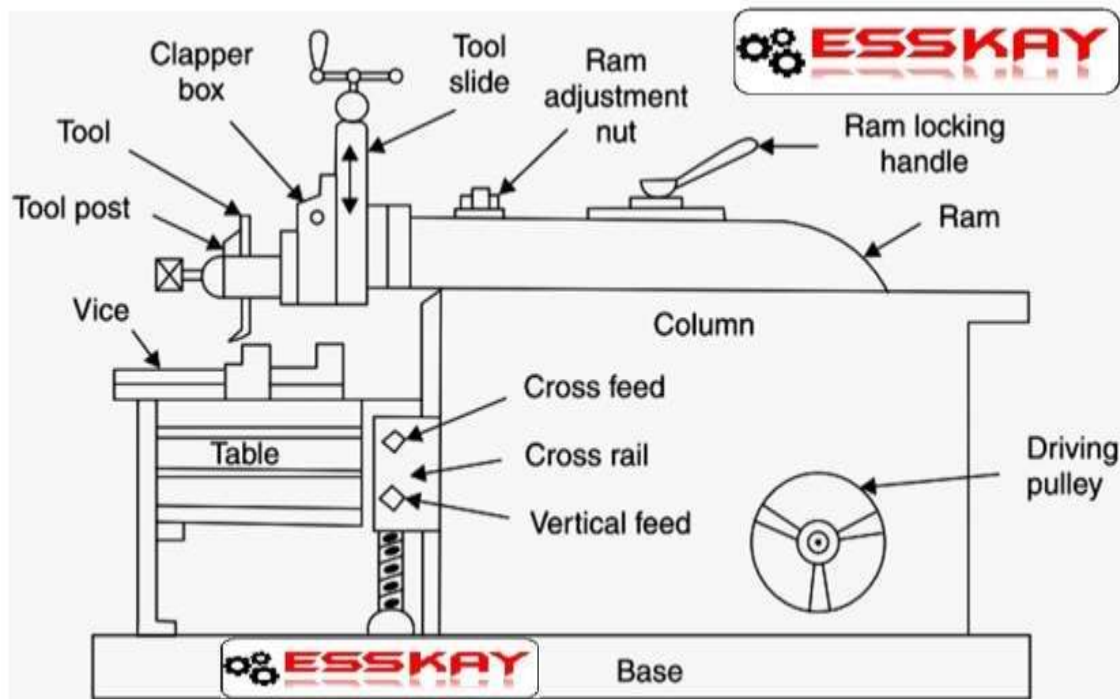
Peripheral or
slab milling



Shaping Machines

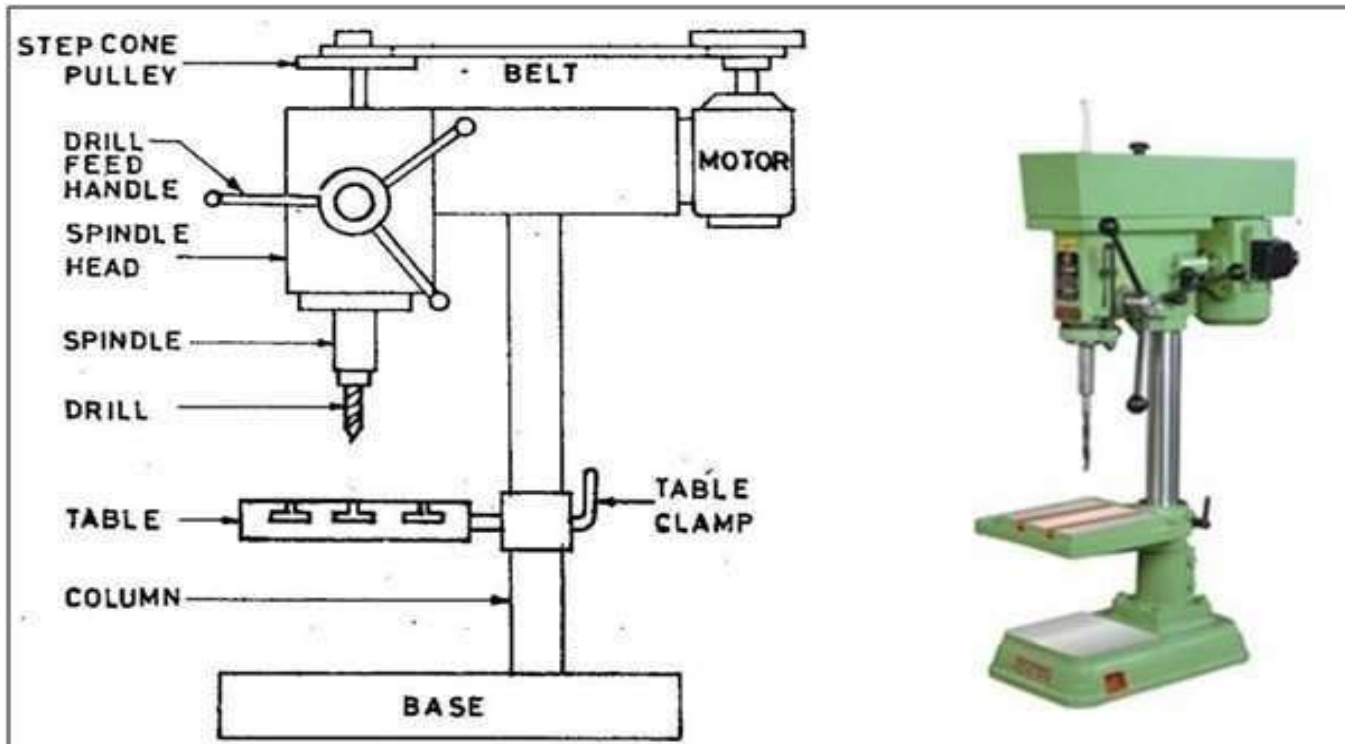
A process in which cutting motion provided by linear-forward motion of the **reciprocating tool** and the intermittent feed motion provided by the slow transverse motion of the workpiece resulting in a flat surface by gradual removal of excess material layer by layer.

Shaping machine is a **machine designed for giving desired shapes to the surfaces** that may be horizontal, vertical and flat. A shaping tool is used to cut in curves, different angles, and many other shapes. A disc is responsible for the tool rotation which results in the forward and backward movement



Drilling machine

- In machining, boring is the **process of enlarging a hole that has already been drilled (or cast)** by means of a single-point cutting tool (or of a boring head containing several such tools), such as in boring a gun barrel or an engine cylinder.
- Boring is a cutting process that involves the use of a single-point cutting tool or **boring head to enlarge an existing hole in a workpiece**. This is in stark contrast to drilling, which is performed to create an initial hole in a workpiece.
- **Boring** is a unit **process** in manufacturing as a mass reduction step, used for enlarging and accurately sized existing hole by means of single point cutting .



Boring Operation

Vertical Boring Machine



Jig Vs Fixture

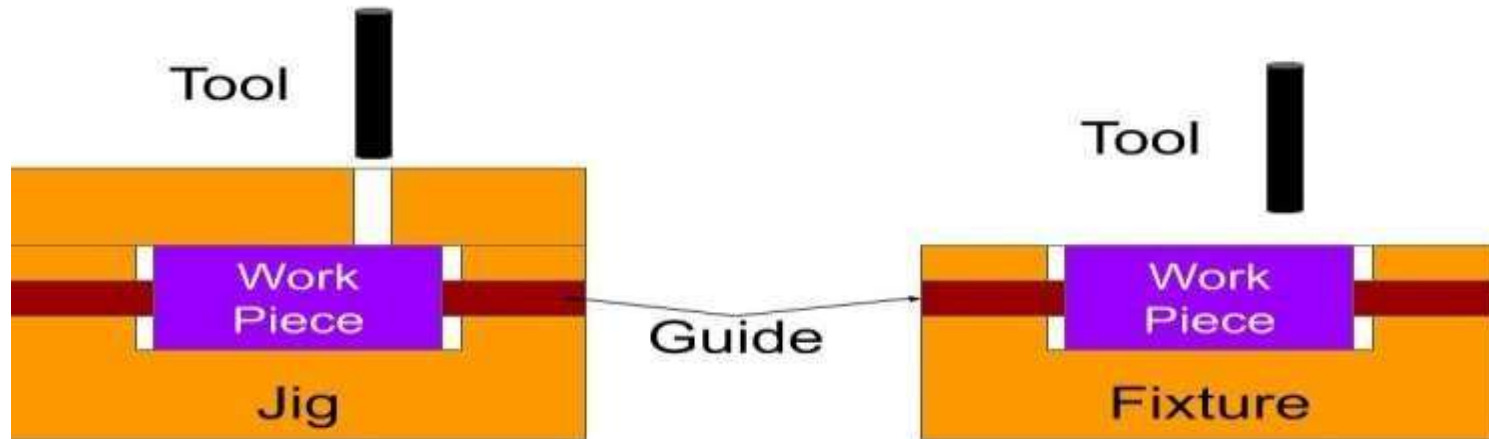
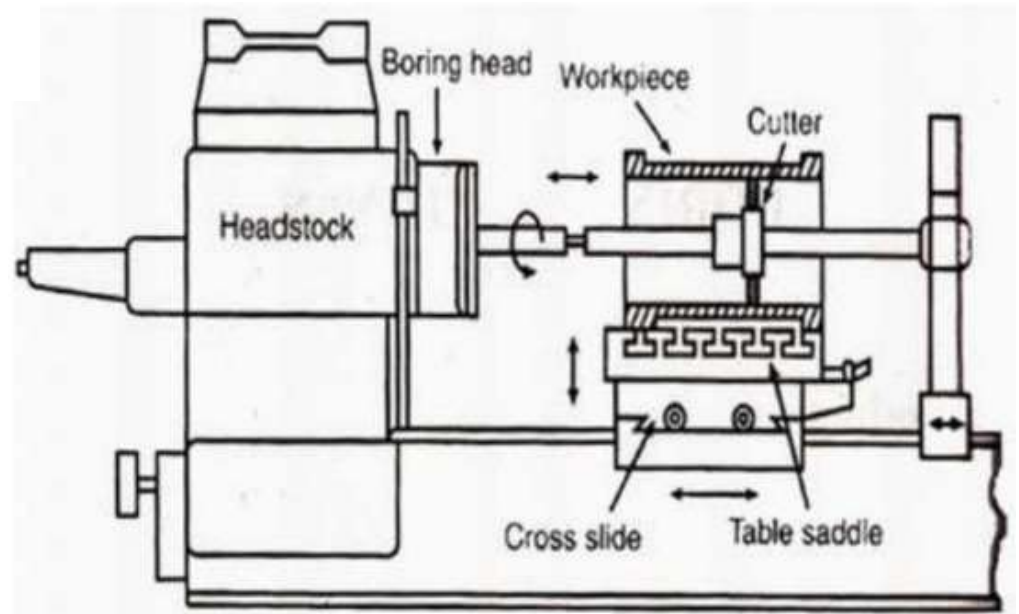


Table type Horizontal Boring Machine

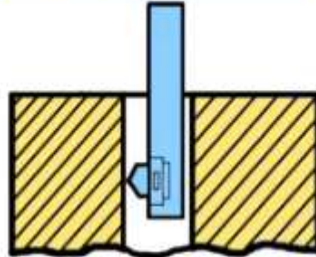


Drilling Operations

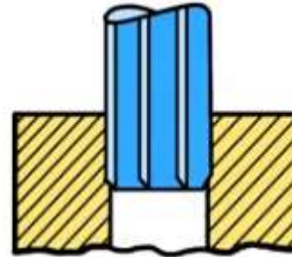
Drilling



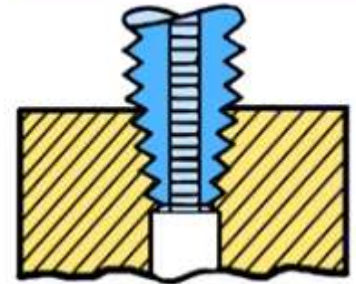
Boring



Reaming



Tapping



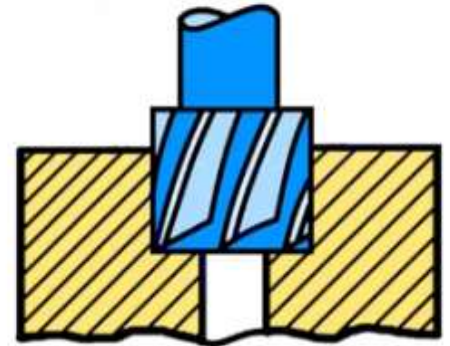
Spot Facing



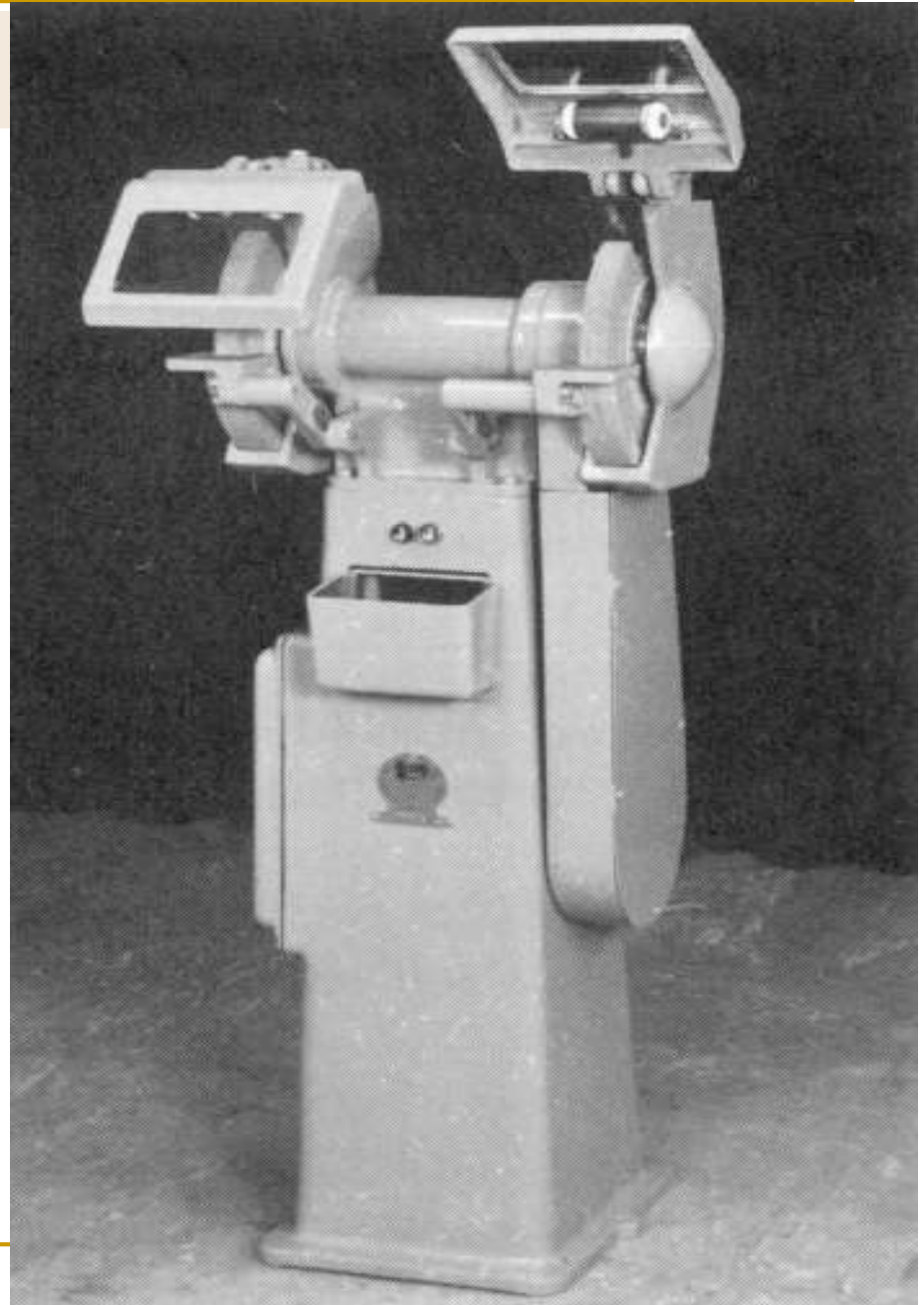
Counter Sinking



Counter Boring



Grinding Machine



Grinding Machine

- The primary purpose of a grinding wheel is to sharpen tools (eg. drill bits).
- The hard abrasive of the wheel is made for removing very hard materials like high speed steel.
 - Never grind on the side of the wheel.
 - Never grind a soft material such as Aluminum. The material will coat the wheel and prevent the abrasive from working properly. (SHOW)

Surface Grinding

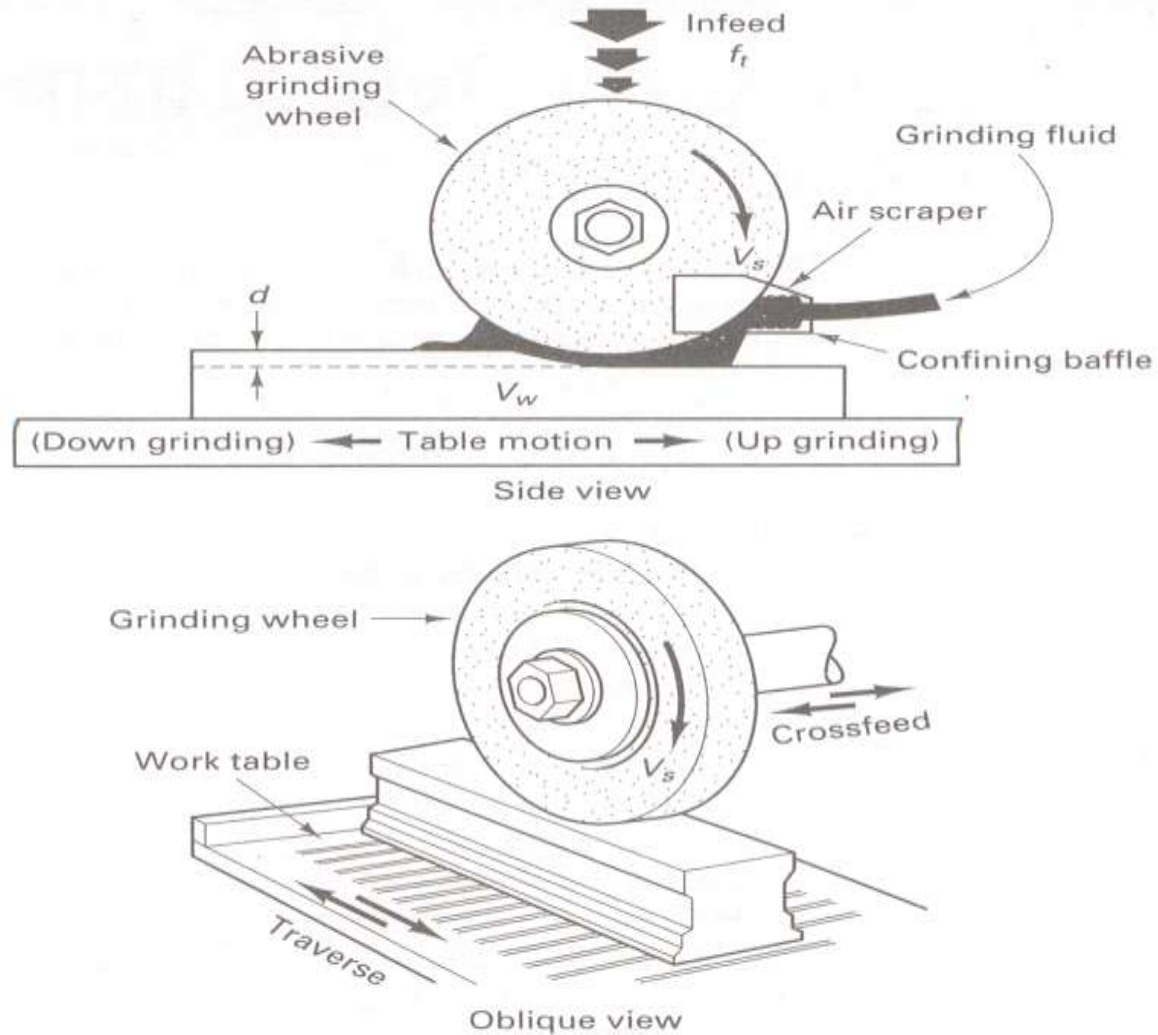
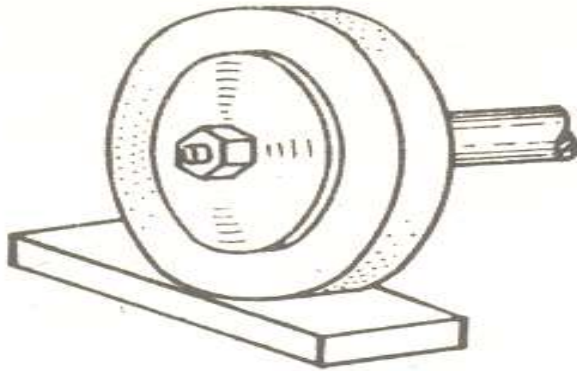
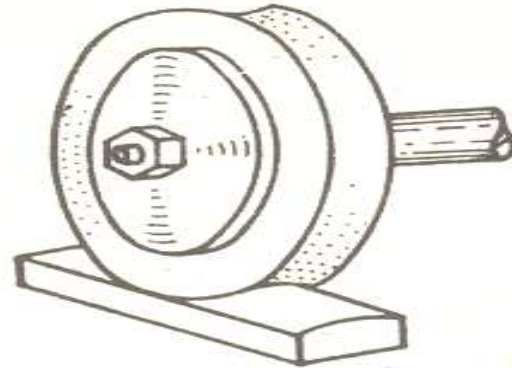


FIGURE 27-1 Schematic of a surface grinding, showing infeed and cross-feed motions along with cutting speeds, V_s , and workpiece velocity, V_w .

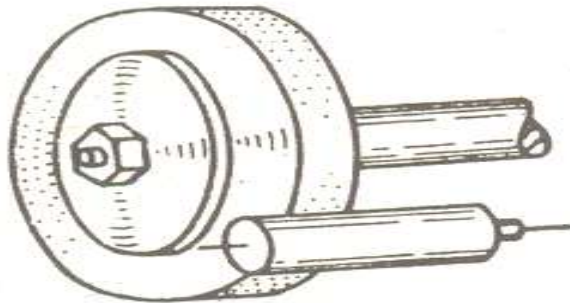
Grinding Operations



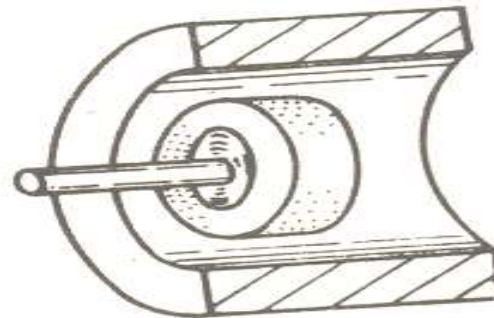
Surface grinding



Form grinding



Cylindrical grinding



Internal cylindrical grinding

Fig. 4.56 Some common grinding operations.

Grinding operations

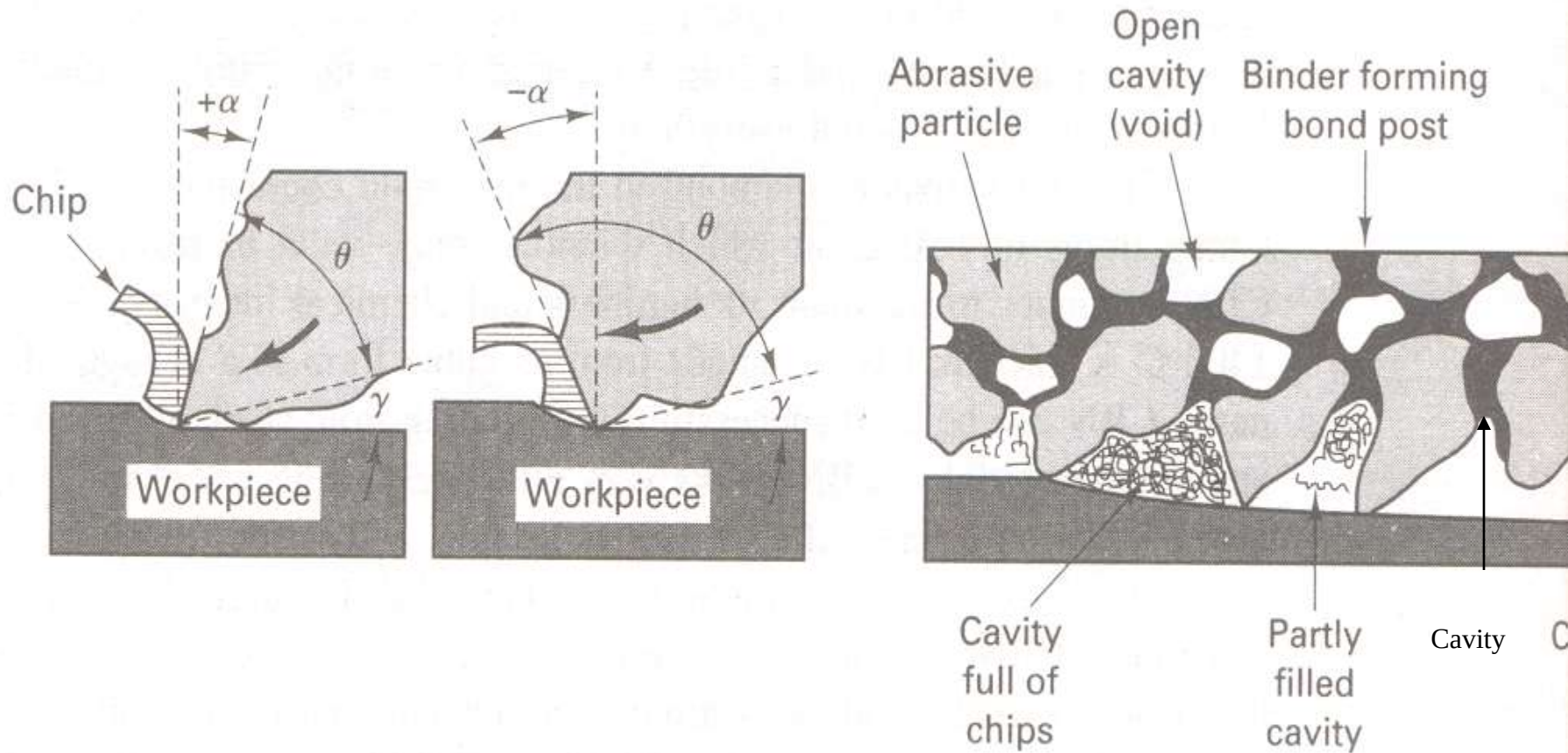
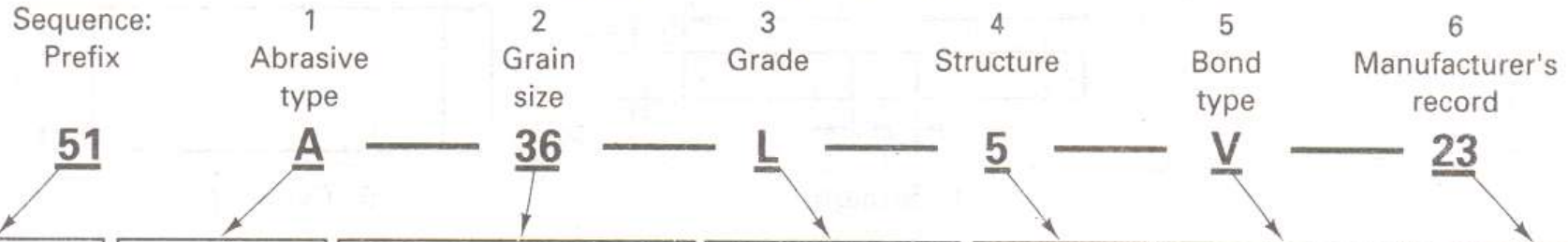


FIGURE 27-4 The rake angle of abrasive particles can be positive, zero, or negative. The cavity or voids between the grains must be large enough to hold all the chips during the cut.

Grinding Wheel Specification



STANDARD MARKING SYSTEM CHART
ANSI STANDARD B74,13 – 1970



PREFIX	ABRASIVE TYPE	ABRASIVE (GRAIN) SIZE				GRADE			STRUCTURE		BOND TYPE	MANUFACTURER'S RECORD
		COARSE	MEDIUM	FINE	VERY FINE	SOFT	MEDIUM	HARD	DENSE TO OPEN			
Manufactuer's symbol indicating exact kind of abrasive (use optional)	A – Aluminum Oxide C – Silicon Oxide	8	30	70	220	A E	I M	Q V	1	9	B – Resinold BF – Resinold reinforced E – Shellac O – Oxychloride R – Rubber RF – Rubber reinforced S – Silicate V – Vitrified	Manufacturer's private marking to identify wheel (use optional)
		10	36	80	240	B F	J N	R W	2	10		
		12	46	90	280	C G	K O	S X	3	11		
		14	54	100	320	D H	L P	T Y	4	12		
		16	60	120	400			U Z	5	13		
		20		150	500				6	14		
		24		180	600				7	15		
									8	16		
									etc. (use optional)			

FIGURE 27-13 Standard marking system for grinding wheels (ANSI standard B74. 13-1970).

Mechanics of Grinding

Mechanics of grinding

- In the analysis of grinding process, all grains are assumed to be identical.
- Two different types of grinding operations namely **plunge grinding and surface grinding** will be considered for analysis.
- In plunge grinding operation a job of rectangular cross-section is being fed radially at the **feed rate f (mm/min)**.
- The uncut chip thickness per grit (t_1) can be expressed as

$$t_1 = f/ZN.$$

Z =number of active grains per revolution in one line

N =rpm of the wheel

Safety Measures

- **Safe Conduct in the Shop**

Be aware of what's going on around you. For example, be careful not to bump into someone while they're cutting with the bandsaw (they could lose a finger!).

- **Machining**

IF YOU DON'T KNOW HOW TO DO SOMETHING, ASK! BEFORE YOU START THE MACHINE:

- Study the machine. Know which parts move, which are stationary, and which are sharp.
- Double check that your work-piece is securely held.
- Remove chuck keys and wrenches.
- DO NOT LEAVE MACHINES RUNNING UNATTENDED!
- CLEAN UP MACHINES AFTER YOU USE THEM!
- A dirty machine is unsafe and uncomfortable to work on.